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MULTI
A Window Based Enviroment Manager

DSM-00083-01 A00

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REVISION	DATE
A00	August 1989
A01	January 1994

MULTI USER'S MANUAL

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CHAPTER 1

INTRODUCTION TO MULTI

Introduction

MULTI is a computer program that creates an environment in which your current software can better match the way you work, without requiring you to replace or make changes in the software you have already purchased.

As important as the computer and the useful software now available can be to your daily business routine, they often do not entirely conform to the way you actually work. MULTI can change this situation, making you and your computer more productive in the process.

Making the Computer Work *With* You

Computers have become an important part of our day-to-day business life. Regardless of whether you are a manager, a secretary, a programmer, or any other type of office professional, you probably depend on the computer for much of your daily work.

But despite this dependence, you probably find that the computer "gets in the way" at times. What should be a simple task takes extra long because it involves doing things the computer's way, instead of yours. Not exactly why you use a computer, is it?

Let's take a look at the type of situations you may run into when using your computer system.

The Manager

If you're a typical manager, your day is divided into seven key activities—planning, managing, telephoning, thinking, numeric analysis, correspondence and meetings—and in theory you are able to allocate the appropriate amount of time to each of those different activities.

In reality, however, no matter how well you plan, phone calls, unscheduled meetings, requests for information, etc., all combine to force those activities to be performed in fragmented segments throughout the day. In order to get anything done at all, you need to be able to switch rapidly from task to task, to take care of interruptions as they occur.

In addition, none of your activities is really independent; the results of a phone call may cause you to change your thinking on calculations you are doing. Furthermore, the results of those calculations often must be included in your correspondence.

The Secretary

As a secretary, you are also responsible for planning, telephoning, correspondence, and meetings. You share many of the frustrations of the manager described above, but are probably interrupted even more often. Answering the telephone can be a source of non-stop interruptions, each requiring a brief, but essential, bit of work—checking a schedule, taking a message or scheduling a meeting—which can seem to waste a whole day.

The Programmer

As a programmer, you not only have the interruptions that we must all cope with, but also have a variety of inter-related tasks to perform as part of your normal workday. There are programs to be designed, source code to be entered, programs to be compiled and debugged, and documentation to be written. Often these tasks could be done in parallel if it were possible: while waiting for a program to compile, you could be editing another source module. In addition, there is a great deal of interdependency among your tasks: a screen display specification that comes from your design document must be included in your program source code, and the resulting display must be included in your final documentation. But despite this interdependency you can feel forced to do things in a step-by-step, sequential order if the computer doesn't let you move easily from task to task.

The Solution: MULTI

While your Alpha Micro computer provides tools to perform each of these tasks—such as AlphaCALC™ worksheet processing, AlphaWRITE™ word processing, AlphaMAIL™ electronic mail, and extensive software development tools—individually these tools do not make up the ideal overall "environment" in which to work. To switch from word processing to worksheet analysis, you must exit AlphaWRITE and enter AlphaCALC—a lot of work if you're only looking for one number to answer a question on the telephone, considering you must now exit AlphaCALC and re-enter AlphaWRITE to get back to where you started. And then to send a quick "while you were out" message after answering the phone, you must once again leave the program you are using,

call up AlphaMAIL, send the message, and start up your original program.

MULTI was created to help solve all of these dilemmas, and more. Known as an "environment manager," MULTI provides access to several different applications at the same time. Without requiring modifications to existing software, and without obsoleting software packages you have already purchased, MULTI "wraps itself around" the software you use, providing the following capabilities:

- You can switch quickly from one task to another, without losing your place or stopping what you were doing. Likewise, you can quickly return to where you started. For example, you can move from an AlphaWRITE document to an AlphaCALC worksheet with two keystrokes, and return just as easily.

- You can easily move data from one application to another even if neither application program was designed to allow data transfer. For example, you can make a copy of a screen display and place it into the user's manual you are producing for your latest software product.

- Without leaving your current computer task, you can quickly call up MULTI's phone list and place a call, or call up the calendar and refer to a particular day's "to-do" list.

- You can use MULTI's built-in calculator to perform quick calculations and then move the result into your document or worksheet.

- Rather than accumulating a pile of small paper scraps on your desk, each containing an important note, you can organize these notes using MULTI's notepad. You can access this notepad no matter what you are doing on the computer, even when in the middle of another program.

- MULTI notifies you of important events via its alarm clock facility.

Here are some typical situations where the capabilities of MULTI might benefit you:

- You are composing a report with AlphaWRITE when you receive a call requesting the final budget figures for next quarter. Rather than having to exit from AlphaWRITE and lose your place, you request that MULTI switch you to your AlphaCALC task, call up the budget worksheet, and read off the requested numbers. After hanging up, a few more keystrokes puts you right back where you were.

- Your associate also asked you to call one of the branch offices to verify they have the same numbers. Not remembering the phone number of that particular office, you quickly call up the phone list accessory, look up the number, and once again return to AlphaWRITE, never having to lose your place.

- You have dialed up an on-line database system to request some current stock quotes. Rather than having to jot all of the numbers down on a piece of paper, simply mark the numbers you want on the screen, move to AlphaWRITE, and copy the information into a document. Or, if you wanted to base some calculations on the stock prices, copy the same information into an AlphaCALC worksheet.

- A colleague stops by your office and requests a listing of one of the program modules you have been working on. Rather than stopping what you are doing, you call up MULTI's printer accessory and request that the listing be sent to the printer, then quickly return to your original task.

- After requesting that a large file be sent to the printer, you want to check on its progress. Without leaving the program you are using, call up MULTI's printer accessory and request a quick look at the printer status.

- You are testing the program you just finished creating and it is asking for the name of your input file—but you don't remember what name you gave to your test data. Call up MULTI's directory accessory for a quick list of your disk files. By scanning through this list you can quickly locate the name of the file you are looking for.

As you can see, MULTI allows you to use the computer more naturally, by allowing it to work *with* you, rather than against you. Instead of forcing you to segment your day into rigid procedures,

or forcing you to suffer the frustrations of not doing so, MULTI allows you to skip back and forth, from one task to the other, incorporating the results from one operation into the next.

Conventions and Symbols

Like other documents in the Alpha Micro documentation library, this book contains a number of standard symbols to make our examples easier to read and understand.

SYMBOL	DESCRIPTION
	This symbol means **STOP!** and signals an important warning or restriction you must know about before continuing.
	This symbol marks a hint, and identifies a shortcut or an easier way to do something.
	This symbol says "Don't forget!" and signals information to remember.
`text`	Text that looks like this in our examples shows the characters the computer displays on your terminal screen such as prompts, information messages, and program samples.
TEXT	Text that looks like this in our examples shows the characters you type on the computer keyboard. For example: **`LOG OPR:`**.
KEY	This symbol tells you when to press the indicated key on your terminal keyboard. For example: **`DIR`**RETURN tells you to press the RETURN key at the end of the DIR command.
KEY/KEY	This symbol tells you to press the first key indicated, and continue holding it down while you press the second key. For example, to type a shifted MENU key, SHIFT/MENU, press the SHIFT key and, while holding it down, press the MENU key.
MENU	The MENU Key. This is a function key used by MULTI and other application programs. On most Alpha Micro terminals the MENU key is labeled F13.

About This Book

There are two parts to this book. The first part (Chapters 1 through 4) introduces you to MULTI, teaches you some of its basic concepts, and gets you running.

The second part (Chapters 5 through 9) teaches you how to tailor MULTI to your particular needs and use some of its more advanced features.

While we have tried to make this manual as easy to use as possible, one of the things that makes MULTI easy to use is its *visual* nature. What appears quite simple and obvious on the screen is often hard to set down in words.

For that reason, always feel free to experiment. MULTI tries very hard to interpret your commands and do what you want, without requiring you to learn a rigid set of rules. Commands are always listed on the screen via menus, and help is always available via the [HELP] key in case you forget how a command works. Or, simply experiment and see what effect each command has.

If you are an experienced AMOS user, or are a system operator attempting to install MULTI for the first time, pay special attention to Appendix A which contains full installation instructions. If you are upgrading from an older version of MULTI to a new one, refer to the *MULTI Release Notes* for important compatibility and upgrade information.

For More Information

If you want to know more about how your Alpha Micro computer operates, and how AMOS (the Alpha Micro Operating System) controls what the computer does, you can read the *Introduction to AMOS* and the *AMOS User's Guide*.

CHAPTER 2

THE MULTI ENVIRONMENT

Introduction

Before we plunge into MULTI, let's take a few minutes to look at some of the basics of using the Alpha Micro computer. If you're already familiar with the operation of the computer and your terminal, you may wish to simply skim this chapter.

In this chapter we'll talk about:

- The terminal and the cursor
- The keyboard
- Logging onto the system

The Terminal Screen

Your terminal screen is your window into the computer. The screen is sometimes called a CRT because it's a Cathode Ray Tube just like your TV set at home. Sometimes the whole terminal is referred to as a CRT or as a VDT, a Video Display Terminal.

The Cursor

Somewhere on your screen there is a small indicator that shows you where the next character you type will appear. This indicator, called the cursor, is usually a small bright rectangle, but may also appear as a small horizontal line. Note that if the cursor is positioned on top of a character on the screen, the character is displayed as dark on a light background, instead of the normal light on a dark background.

The Keyboard

The keyboard of your terminal resembles the standard typewriter keyboard. In addition to the keys you are familiar with, there may be several extra keys and perhaps a group of keys arranged like a ten-key adding machine off to the right-hand side. The instruction manual that came with the terminal can give you a lot of information about your terminal and how it may be used with the various programs on your computer other than MULTI.

The RUBOUT Key

If you make a mistake, the RUBOUT key lets you backspace the cursor and erase at the same time. It backspaces and deletes one character each time you press it.

The Arrow Keys

To one side of the main keyboard you'll see four keys with arrows that each point a different direction; these are known as the ARROW or CURSOR keys. These keys are specifically designed for moving the cursor around the screen.

Other Special Keys

Near the arrow keys are special keys with labels such as INS LINE, HOME, etc. Each of these keys tells MULTI to perform a specific function. Each of these keys performs two different functions, depending on whether the SHIFT key is pressed at the same time. (We will indicate the use of the SHIFT key in this manual with the symbol [SHIFT].) The function of these keys will be discussed later in this manual.

Function Keys

There is a fourth group of keys above or to the side of the main keyboard. These are known as function keys and are typically labeled something like [F1] through [F16]. In many applications, such as AlphaWRITE and AlphaCALC, these function keys are labeled by means of a plastic overlay. MULTI lists the function keys to use in its menus.

Like the keys around the arrow keys, function keys may be used with or without the [SHIFT] key, allowing each key to perform two functions.

The Shifted MENU Key

One key is particularly important to the operation of MULTI: the shifted MENU key. This key is labeled on the AlphaWRITE and AlphaCALC plastic overlays, and is normally assigned to function key [F13]. By pressing [SHIFT] and [MENU] together, you can get MULTI's attention. For instance, when you are in AlphaCALC and wish to switch to AlphaWRITE, you first press [SHIFT] and continue holding it down while pressing the [MENU] key. Even though you have not exited AlphaCALC, MULTI will now display a command menu, from which you can switch to AlphaWRITE. We

will indicate the use of the shifted [MENU] key by the symbol [SHIFT]/[MENU]. If your terminal does not have a [MENU] key, you can assign the [SHIFT]/[MENU] function to another key; see Appendix A for more information.

Universal Key Sequences

If your terminal doesn't have all the keys MULTI uses, you can still use all of MULTI's functions by using the universal key sequences defined in Appendix B.

Logging onto the Computer

Before you can actually use the computer, you have to ask your System Operator to assign you an account number. The account number comes in two parts: the Device Name, and the Project-Programmer Number.

The Device Name identifies the disk the account is located on. It is always followed by a colon, so your assigned device might have a name like DSK1: or SUB2:.

The Project-Programmer Number is the computer's way of keeping track of the data on a disk. It is always enclosed by brackets, and the project number and programmer number are always separated by a comma. Yours might look like this: [100,5] or [34,2].

Now you're ready to log on. First, make sure the System Operator has the computer system completely powered up and ready to go. The cursor should be waiting next to the AMOS prompt. Enter the word LOG followed by your account number like this:

`LOG DSK2:[34,2]` [RETURN]

The [RETURN] symbol indicates you should press the RETURN key when you finish typing the line. (Depending on your particular terminal, the RETURN key might be labeled RETURN, ENTER, or be marked with a left-curving arrow.) The next thing you should see on your screen is the message:

```
Logged into DSK2:[34,2]
```

Now you're ready to begin learning how to use MULTI and seeing how it can ease your everyday use of the computer. The next chapter teaches you some of the fundamental concepts used by MULTI.

CHAPTER 3

BASIC CONCEPTS

Introduction

This chapter discusses some of the basic concepts used by MULTI with which you will need to be familiar. You will learn about:

- The MULTI Menu
- Various methods of entering commands
- How to get help

The Basics

MULTI lets you perform multiple functions from your terminal at the same time by allowing you to control multiple *windows*. Each window is a separate *task*, just as if you have added a new terminal to your computer. For this reason, each window can execute a different function. You can open and close windows as you need them, and switch the functions they perform, through commands to MULTI.

The same restrictions that apply to multiple users on the Alpha Micro computer also apply to windows. For example, just as two different users can't edit the same file at the same time, you can't edit a file in one window and, without exiting the file, also edit the same file from another window. Such restrictions serve to protect disk files from being damaged.

The Current Window

At any time, one of your windows will be displayed on your screen just as if MULTI was not being used. However, MULTI is still keeping track of what is happening on all windows, so that you can switch from one window to another.

The window that you see on your screen is considered the *current window*. Any other windows you have open are also running and being updated—you just can't see them. Selecting another window as the current window will allow you to see what is happening.

You may interact with the current window just as you would interact with the computer outside of MULTI. When you enter characters from the keyboard they will be sent to this window, and any responses will be displayed. If you are already familiar with using your Alpha Micro computer, then you are already familiar with MULTI. You will not even be aware that MULTI is present until you call upon one of its features. Certain combinations of terminals and software, however, do slightly alter the appearance of the screen due to terminal hardware limitations. See Appendix D for more details.

If you are just getting started with your Alpha Micro computer—not to worry—all of the manuals and tutorials describing the system fully apply when you are running with MULTI.

The name of the current window will appear on the top status line of your terminal if the MULTI initialization file is set up to allow it (see Appendix A for details). The name of the current window will always appear on the bottom status line whenever the MULTI Menu is visible.

The MULTI Menu

To allow you to perform other functions, MULTI provides a *menu* of commands from which you can choose. This menu appears as a box along one side of your terminal display and is labeled "MULTI."

Whenever you wish to see the menu so you can give commands to MULTI, press [SHIFT] and [MENU] at the same time. The menu will appear and you may enter your commands as described in the next section.

When you wish to leave the MULTI Menu and return to where you were previously, press [SHIFT]/[MENU] again. The menu will disappear and you may continue working.

Giving Commands to MULTI

All commands directed to MULTI are entered from the MULTI Menu. As each command is performed, you may be asked for additional information.

Several different methods are provided for entering commands so you may choose the method you find most comfortable. Although much research has been done to try to determine the best method of entering commands, it still comes down to a matter of personal preference. Often, you will find yourself using a variety of command entry methods as you evolve your own style of using MULTI.

You may enter all commands directly into the MULTI Menu by typing the command name followed by pressing [RETURN]. Command names may be abbreviated to as few characters as makes them unique.

Commands may also be entered by pressing the appropriate function key, as shown on the MULTI Menu, or by using the universal key sequence equivalent to the function key.

Regardless of how you enter commands, the results are always the same.

Getting Help

Whenever MULTI is asking for input, such as when the MULTI Menu is visible, or when some other question is being asked, you may request further information by pressing the [HELP] key.

Upon pressing the [HELP] key, a window opens in the lower half of your screen, and additional information appears within it. After you have read the information, press any key to return to the point where you pressed [HELP]. If you wish to terminate the command without entering any response (such as when you discover you have entered the wrong command), press [SHIFT]/[MENU] to return to the MULTI Menu.

CHAPTER 4

GETTING STARTED WITH MULTI

Introduction

Now that you know how to use your terminal and keyboard, and understand the basic concepts behind MULTI, we can begin with a simple example to teach you the basics of MULTI.

The topics covered will be:

- Starting the MULTI program
- The OPEN Window Command
- The QUIT Command

Starting MULTI

Before trying to use MULTI, make sure your computer has been properly configured. Contact your System Operator to ensure MULTI has been installed according to the instructions in Appendix A, and you have sufficient memory.

Set aside a particular disk account that you always log into before bringing up MULTI. (Once you open a MULTI window, you can then log that window into whatever disk account you want.) MULTI creates and maintains certain files, such as your phone number lists and your notecards, in the account you logged into before calling up MULTI. You will want to make sure this disk account is used by you alone.

Once you are sure everything has been set up correctly for you, you are ready to enter MULTI. All you have to do to get into MULTI is, type:

MULTI RETURN

Try this out on your terminal. When you see the MULTI Menu on your screen, you are inside the MULTI system. At this point there are no windows open, so most of your screen is blank. Later, we will show you how to have MULTI automatically open windows for you each time you enter MULTI.

The OPEN Command

To create a window, use the OPEN command. Type **OPEN** or **O**; then press RETURN. Or, press F1. A command window appears and MULTI asks you for a name for your new window.

Your cursor is at "Enter window name:". Here is where you type the name you wish to give to your new window.

The window name can be up to fifteen characters long. You can include any characters you wish in the name, including spaces. You may type the window name in either upper or lower case, and MULTI always displays the window name exactly as you typed it. However, MULTI doesn't care whether you type a window name in upper or lower case. This makes it easier to type in window names when when you select a window since you don't have to worry about matching the exact case of the name.

It is a good idea to try to make the window name something that describes what you intend to use the window for, so it is easy to remember as you switch from one window to another.

Type a window name now, such as **Demo**, and press RETURN. MULTI now asks several questions allowing you to define the memory allocation and the application you wish to run in the window. We will discuss these questions in the following text.

To save some time, MULTI allows you to specify the name of the window you wish to create on the same command line where you specified the OPEN command. Thus, to create our window called Demo we could have typed **OPEN Demo** RETURN, and MULTI would not have needed to ask for a name. This feature of MULTI, which is included in many commands, is designed solely to save you typing time; the results of the command are identical, regardless of which method you use for selecting the name.

Application

After you have typed the window name, MULTI asks what you wish to do within the new window. Although you have complete freedom to do anything you wish within the window, it is often convenient to have MULTI automatically start a particular application within a window. This will be useful later, when you learn to define the arrangement of windows MULTI comes up with.

You may select any of the applications listed on the menu, providing it is available on your computer, by pressing the function key listed next to the application you want to use.

For now, press [F6], selecting AMOS command level.

Memory Partition Size

To be able to use the window you are creating, you must assign it some memory. After specifying the application you wish to run in your window, MULTI asks for the amount of memory you wish to assign to it.

Several choices appear in the command window; these include the default value for the application you have selected (as choice [F1], an opportunity for you to specify an exact amount of memory (as choice [F2]), and several other choices depending on the amount of memory your own job has been assigned on the computer. Only those choices actually available to you, based on the approximate amount of contiguous memory you have and the application you have chosen to run, are displayed.

The amount of memory you wish to assign to your window depends on what it is you wish to do later within the window. For now, select the default value of 64K by pressing the [F1] key.

You cannot select a window memory size of less than 2K.

Using Your Window

After you choose a memory allocation, your have a fully functional window ready to accept your commands. The window is automatically logged in to the same device and account you were in when you first called up MULTI.

Any command you type on the keyboard is sent to your window to be executed. For example, call up the SYSTAT command by typing **SYSTAT** and pressing [RETURN].

Note how each character you type is echoed within your window. Likewise, once you press [RETURN], the SYSTAT display is shown as it would be without MULTI.

Leaving MULTI

To complete the demonstration of the basics of MULTI, we will show you how to leave MULTI. First, call up the MULTI Menu by pressing [SHIFT]/[MENU]. To leave the menu and return to the AMOS command level, type the command **QUIT**[RETURN] or **Q**[RETURN] or press [SHIFT]/[F9].

Whenever you have any windows open, as you do now, MULTI asks for confirmation before allowing you to quit. This is done to avoid losing data if you should accidentally type "QUIT" or press the quit function key. When asked for confirmation, type **Y** to confirm you do indeed wish to quit. If no windows are open, MULTI will not ask for confirmation.

Normally you see a slight pause and then return to AMOS command level. If, however, you are in the middle of editing a document or performing some other task within your window, MULTI does not let you quit. This is to prevent you from inadvertently leaving MULTI before completing some task within a window. As we will see later on, when many windows are active, it is very easy to forget exactly what is going on in each window.

If MULTI decides it cannot let you exit, it tells you which window is causing the problem at the bottom of the screen. Select that window to determine why MULTI will not let you exit, and return the task within that window to AMOS command level.

Typically, MULTI lets you quit as long as each window is in one of the following states:

- AMOS command level;
- The outer menu of AlphaCALC, AlphaWRITE, or AlphaMAIL with no text on the command line;
- The outer menu of AlphaMENU;
- Interactive mode of AlphaBASIC;
- A program that you can exit by pressing Control-C;
- A program which responds to an SI$EXI software interrupt.

What's Next?

This chapter has shown you the basic operation of creating and using a single window. Of course, much of the power of MULTI is that it lets you use many different windows at the same time.

In the following chapters we will see how we can use these multiple windows, in combination with some new commands we will learn about, to perform many other operations.

CHAPTER 5

USING MULTIPLE WINDOWS

Introduction

This chapter discusses the techniques you can use to manage multiple windows at one time, and how to switch back and forth between them. You will learn about:

- Opening and closing windows
- Selecting and using the current window
- Switching from one window to another

Opening Multiple Windows

In the last chapter we learned how to open a single window. As we mentioned, however, most of the power of MULTI stems from the ability to have several windows open at the same time. It is this ability which allows us to quickly transfer between different applications and, as we'll see later, transfer data from one window to another.

To open more than one window at the same time, repeat the use of the OPEN command we learned about in the last chapter. You can have up to 16 windows open at the same time, providing you have enough memory to assign to them. However, in practice more than 3 or 4 windows open at the same time usually ends up being more confusing than it is helpful. Experiment to determine the right number of windows for your style of working.

Closing Windows

When you are done using a window you can *close* it. Once a window is closed, its contents are lost and cannot be recovered. Closing a window also makes the memory assigned to the window available to any new windows you might open.

To close a window, call up the MULTI Menu. Type **CLOSE** or **C** followed by the name of the window you wish to close and press RETURN. For example, to close a window named DEMO, type **CLOSE DEMO** and press RETURN. Or, press F2 to see a list of windows from which to choose, and

press the function key listed next to the name of the window you want to close.

If you don't specify the name of the window you want to close, MULTI asks you for it.

After pressing RETURN, you normally see a slight pause and the screen then redisplays without the closed window. If, however, you are in the middle of editing a document or performing some other task within the window you try to close, MULTI will not close the window. This is to prevent you from inadvertently closing a window before you are done with it, thus losing valuable work.

If MULTI decides it cannot close a window, it tells you so at the bottom of the screen. Use the FULL command to see what is going on in the window then return the task within that window to AMOS command level.

Typically, MULTI will not let you close any window that is *not* in one of the following states:

- AMOS command level;
- The outer menu of AlphaCALC, AlphaWRITE, or AlphaMAIL with no text on the command line;
- The outer menu of AlphaMENU;
- A program you can exit by pressing Control-C;
- AlphaBASIC interactive mode;
- A program which responds to the SI$EXI software interrupt.

Selecting a Window

As we discussed earlier, one window is always designated the *current window*.

Except when the MULTI Menu is visible, or when using one of MULTI's desk accessories, anything you type on your terminal keyboard is sent to the current window.

If MULTI is using the top status line (that is, it is showing the current time and date at the top of your terminal screen), it also shows you the name of the current window. If you have selected the top status line for use by your application program, as discussed in the next chapter, you can determine the name of the current window by calling up the MULTI Menu and looking at the bottom line of the screen.

How to Select a Window

You may select a window as the current window by using [SHIFT]/[MENU] to bring up the MULTI Menu, and:

1. Typing the name of the window you wish to make current;

2. Or, typing the command **SELECT** or **S**, and pressing [RETURN]. Then you see a list of windows from which to choose. Press the function key listed next to the window you want to select.

3. Or, pressing [F7]. You then see a list of windows from which to choose. Press the appropriate function key.

Switching Between Windows

A quicker method of moving between windows is to use the *Swap Windows* key. When you press this key MULTI automatically chooses the next window (in the order in which you opened them) and selects it as the current window. Successive uses of the Swap Windows key continue to cycle through all open windows. The Swap Windows key, therefore, provides a very quick way to move between windows as it does not require you to call up the MULTI Menu.

For example, if you are working with two windows, say a word processor and a spreadsheet, a single keystroke toggles you back and forth between those windows. Even when using more windows you may find it quicker to cycle through the windows with a series of single keystroke commands than calling up the MULTI Menu and selecting the desired window by name.

If you are using an Alpha Micro terminal, the keystroke to switch windows uses the [FUNCT] and [S] keys. On an AM-60, press the [FUNCT] key, release it, and then press [S]; on an AM-62, AM-62A, AM-65, AM-70, AM-72, or AM-75, hold down on the [FUNCT] key while pressing [S].

On other terminals you can assign the Swap Windows function to any keystroke you wish via the SWAPKEY command in the MULTI.INI file. See Appendix A for more information on this option.

Quick-Selecting a Window

The very quickest method of moving between windows is to use the *Select Window* key. You press this key, then press another key from A to Z to move directly to the desired window. This makes any window only two keystrokes away.

For example, if you are working with three windows, two keystrokes will move you directly to your window of choice. Press the Select Window key. Next, press a key, from [A] to [P], to select the desired window.

Pressing [A] will select the first window, [B] will select the second, etc.

If you are using an Alpha Micro terminal, the keystroke to switch windows uses the [FUNCT] and [Z] keys. On an AM-60, press the [FUNCT] key, release it, and then press [Z]; on an AM-62, AM-62A, AM-65, AM-70, AM-72, or AM-75, hold down on the [FUNCT] key while pressing [Z].

On other terminals you can assign the Select Window function to any keystroke you wish via the SELECTKEY command in the MULTI.INI file. See Appendix A for more information on this option.

CHAPTER 6

SETTING UP MULTI FOR AUTOMATIC STARTUP

Introduction

This chapter discusses how you can define window configurations which will be automatically set up each time you use MULTI. You will learn about:

- Saving MULTI configurations
- Requesting special startup features using the MULTI.INI initialization file

MULTI always maintains a record of the *current configuration*. This record contains the names, positions, shapes, colors, and selected applications of all windows currently open. The application stored is the one selected when you first opened the window, and not necessarily the application currently running in the window.

MULTI maintains this information so if you save the current configuration you can later return to the same arrangement of windows. Note that saving the window configuration does not save the *contents* of a window, just its position, color, size, etc.

Saving the Current Configuration

Once you have an arrangement of windows you like, and would like to be able to call up again later, you will want to save the configuration. To do so, enter the MULTI Menu and type **SETUP** [RETURN] or **SET** [RETURN] or press [F8]. A menu then appears giving you a choice of several setup functions. Choose the "save current window configuration" option by pressing [F1].

Because you may want to save several different window configurations, MULTI will ask you for a filename under which to save the window configuration. You can choose the default filename MULTI by pressing [RETURN] when asked for the filename. This default filename is the one MULTI will look for each time you call up MULTI. If you wish to save the current configuration under a different name, type the filename (without

an extension) and press RETURN. The filename can be from one to six letters or numbers, with upper and lower case letters being treated the same. MULTI will always use the extension .WSV when storing the file.

Before you type the name you want to use for the configuration file, you might want to use MULTI's directory accessory to verify no configuration file already exists using that name, since MULTI will overwrite any existing file using the same name.

Using a Configuration File

Once you have saved a configuration with the default name via the SETUP SAVE command, MULTI will use that configuration file each time you call up MULTI from AMOS command level. MULTI will automatically open and position windows just as they were when you saved the configuration.

When you call up MULTI by typing **MULTI** RETURN, MULTI looks for the default configuration file (MULTI.WSV) created when you use the SETUP SAVE command with the default filename.

If you specified a filename when you saved the window configuration, you can specify that filename when you invoke MULTI. That is, to call up MULTI and use the configuration file named PROG, type the command:

MULTI PROG RETURN

If you wish to invoke MULTI without using the MULTI.WSV configuration file, type the command:

MULTI/NOSTART RETURN

If you call up MULTI without specifying a saved configuration file and it is not able to locate the default file MULTI.WSV, or a MULTI.WSV file exists but you have used the /NOSTART option, MULTI will start up with no windows open, giving you the MULTI Menu.

Starting Up in a Specific Window

Once you have used the SETUP SAVE command to save a configuration file, you can ask MULTI to start up with one of the saved windows as the current window by typing:

MULTI/FULL:window-name RETURN

where you have a MULTI.WSV file and ***window-name*** is the name of the window you want to start up in. The window you select must be one of the windows previously saved in the configuration file.

If you want to use the /FULL option with a configuration file other than MULTI.WSV, include the name of the configuration file before the /FULL option. For example, to start up in window TEST which was saved in the configuration file WIN1, type:

```
MULTI WIN1/FULL:TEST RETURN
```

The MULTI.INI initialization file (discussed later in this chapter) also allows you to use a FULL command to start up MULTI with a specific window. If the MULTI.INI file exists and contains a FULL command, that command overrides the use of a /FULL: option when you first bring up MULTI.

Having MULTI Start Your Application

Whenever you open a window, you must specify an application you wish to have run within that window. MULTI saves this information and will also start the specified application running when the window is opened as the result of reading a saved configuration.

Remember, even though you may leave the application you specified when you opened the window and go on to another application, MULTI does not know this and will always start up the application specified during the OPEN command.

When you use the OPEN command, MULTI gives you the choice of selecting several common applications. In addition, it gives you a choice of "Other." By selecting "Other" you can specify to MULTI the command line, of up to 60 characters, that you wish to have the window execute each time it is opened. You can also change this list of applications by using the SETUP command—see Appendix A for details.

By having this command line invoke a command or .DO file, you can have MULTI start up just about any application in your window.

The MULTI.INI Initialization File

In addition to reading a saved window configuration file, MULTI also scans the file MULTI.INI each time it is started. You can embed commands in MULTI.INI which can affect the way MULTI starts up and behaves once running.

MULTI.INI is an ASCII text file you can create and edit using the AlphaVUE text editor. In order to be as flexible as possible, MULTI goes through a fairly involved search procedure when scanning for the MULTI.INI. This allows you to use one or more MULTI.INI files to customize the operation of MULTI for each user, or for each group of users.

Each time it starts, MULTI scans for a file called MULTI.INI first in the account you are logged into. If this file is not found, MULTI looks in your project library account ([p,0]). If it is still not found, MULTI looks in the

ersatz device MULTI:. If no file has been found at this point, MULTI goes through the same search path again, but looks for a file with the name MULTI.*lan* where "*lan*" is the language dependent extension defined in the language definition file currently selected for your job (for example, MULTI.USA, when American english is the selected language).

Regardless of whether MULTI has found a MULTI.INI file during the search procedure described above, MULTI always looks for a file called *term*.INI in MULTI:, where *term* is the name of the terminal driver in use by your job. This allows you to place special MULTI.INI commands which are terminal-type dependent in this file.

All commands in the MULTI.INI file have the format of a command followed by one or more arguments. The available commands are described below. Additional information can also be found in Appendix A.

The SMALLCALC Command

MULTI's desk calculator accessory, as described in Chapter 8, normally displays a picture of a calculator on your screen. This picture helps serve as a reminder of the functions that are available. However, this picture also obscures one corner of the screen. To allow you to use the calculator without covering up important data on the screen, MULTI provides a "small" mode for the calculator, which displays only the result of your calculations, thereby reducing the amount of screen area covered.

This "small" mode can be specified any time you invoke the calculator accessory or, if you want the calculator to always display in "small" mode without having to type extra characters when you invoke it, you can use the SMALLCALC command within your MULTI.INI file. For example:

```
SMALLCALC=TRUE
```

The WORDWRAP Command

MULTI's notepad accessory, as described in Chapter 8, normally wraps text at the end of each line, as you enter it. This is known as "wordwrap" and helps you enter text without worrying about pressing RETURN at the end of each line.

However, some people prefer not to have wordwrap active. If you want to disable this feature, you can use the WORDWRAP command within your MULTI.INI file. For example:

```
WORDWRAP=FALSE
```

The FULL Command

When MULTI first is called up, it selects the first window it opens as the current window unless the /FULL: option was used on the MULTI command line. If you want to, however, you can start in a particular window by including the FULL command within your MULTI.INI file.

When you use the FULL command you must specify the name of the window you wish to select as the current window. Make sure you specify a window name that is included in your saved window configuration file. The format of the FULL command is:

```
FULL=window-name
```

where `window-name` is one of the windows saved in the configuration file.

The PRINTER Command

The PRINTER command allows you to define the printers available to MULTI. This command is discussed in detail in Appendix A.

The MASTERMENU Command

The MASTERMENU command allows you to redefine the key used as the shifted [MENU] key to call up the MULTI Menu for use on non-Alpha Micro terminals. This command is discussed in detail in Appendix A.

The SWAPKEY Command

The SWAPKEY command allows you to redefine the key used as the SWAP WINDOWS command key for use on non-Alpha Micro terminals. This command is discussed in detail in Appendix A.

The SELECTKEY Command

The SELECTKEY command allows you to redefine the key used as the Select Window command key for use on non-Alpha Micro terminals. This command is discussed in detail in Appendix A.

The TOPSTATUS Command

Normally MULTI uses the top status line of your terminal to display the current time and date, as well as the name of the currently selected window. Because some applications display important information on the top status line, MULTI provides a way to return control of the top status line to the application program.

By including a `TOPSTATUS=TRUE` command within the MULTI.INI file, the top status line will be available for use by your programs.

CHAPTER 7

TRANSFERRING DATA BETWEEN WINDOWS

Introduction

One of the most important features of MULTI is its ability to transfer data from one application to another, even if no compatible interface exists between the applications. This chapter discusses the techniques you can use to effect this transfer. You will learn about:

- Selecting the data to transfer
- Copying the data

Transferring Data Via the Terminal Screen

MULTI allows you to transfer data between applications and between windows without requiring any special interface to the application software. In many cases, this transfer is done using software that traditionally has no way of copying data in or out.

For example, a terminal emulator program, such as VersiComm, may allow you to have access to an on-line database system containing current stock quotes. Typically, such systems provide no way to access this data other than by displaying it on your terminal. With MULTI, you can take this information from your terminal screen and copy it to another application, such as AlphaCALC or AlphaWRITE. This capability gives you many of the advantages of "integrated" software, without having to modify or give up existing applications.

MULTI allows you to transfer up to a full screen of information from one application to another. While you will usually copy data from one window to another, you can, in fact, copy it to a different location within the same window.

Selecting Data to be Copied

The first step in copying data is to select the data to be copied. MULTI allows you to copy any rectangular area of the screen, up to the full screen. To copy data that does not fit on a single screen, you must select and copy multiple sections.

The CUT Command

You select the data to be copied from the current window. Do this by selecting the window you wish to cut data from and calling up the MULTI Menu. Type the command **CUT** RETURN or **CU** RETURN or pressing F4.

A small arrow appears in the middle of the screen. You can move this marker using the arrow keys. Select two diagonal corners of the rectangular area you wish to copy. This is done by moving the arrow to the first corner and pressing RETURN, then moving the arrow to the opposite diagonal corner and pressing RETURN again. You can mark anything from a single character to an entire screen in this manner.

To make moving the arrow easier and quicker, you can use the HOME key to move it to the upper left corner of the screen; SHIFT/HOME to move it to the lower right corner of the screen; SHIFT/← to move it to the far left side of the screen; or SHIFT/→ key to move it to the far right side.

After selecting the second corner, the selected rectangle redisplays in "dim" mode, to show you the area you have selected. The selected area is then copied to a special area inside MULTI referred to as the *scrap area*.

Quite often you wish to cut the entire screen image. You may do this by selecting the upper left and lower right corners, using the commands described above, or you may take a shortcut by simply pressing S (for "screen") when you are prompted for the first corner. The entire screen is then selected and copied to the scrap area except for the status lines.

Other times you want to include the top and bottom status lines in your selection, such as when preparing documentation based on a screen display. MULTI does not allow you to move the selection arrow into the status lines. However, pressing T (for "total screen") selects the entire screen, including the top and bottom status lines. (You won't get the top status line if your MULTI.INI file does not contain the line `TOPSTATUS=TRUE`.)

If you make a mistake in selecting the area to be copied, use the CUT command again and specify a new rectangle.

There is only one scrap area. Each area you cut and copy to the scrap area overlays the previous contents of the scrap area. Anything you cut to the scrap area stays there until you overlay it with something new or you exit from MULTI. You can therefore cut data *to* the scrap area once and then paste it *from* the scrap area as many times as you wish.

Note that the "dimming" of the selected rectangle is only temporary and that any action which causes the screen to be redisplayed removes the "dimming."

Copying the Data

Once you have placed data into the scrap area via the CUT command, you are ready to use it. Once data is in the scrap area, you may use it as many times as you wish, even pasting it to several different windows, without having to go back and re-mark it.

There are two operations you can perform on the selected scrap: you may paste it to another place, either in the same or a different window, or to a disk file; or, you can print the scrap on one of the system printers.

The PASTE Command

To copy data from the scrap area, use the PASTE command. Position the cursor to the place within a full screen display where you wish the scrap area copied to. Select the MULTI Menu by pressing [SHIFT]/[MENU], and typing the command **PASTE**[RETURN] or **PA**[RETURN], or pressing [F5].

The contents of the scrap area now go to the application running in the current window, just as if it had been typed in on your keyboard.

MULTI attempts to match its method of copying data to the running application. For example, when copying data to an Alpha-CALC spreadsheet, it preserves numeric formatting of data by properly specifying the editing mode for each worksheet cell copied.

It is possible, however, to copy inappropriate data to an application via the PASTE command. For instance, it is possible to copy alphabetic characters to a program which only accepts numbers. The exact result, of course, depends on the particular application program the data is being copied in to. For this reason, make sure the data you are copying is appropriate to the application before using the PASTE command, and that the application is set up correctly to receive the data.

The PRINT Command

In addition to copying the contents of the scrap area, MULTI also gives you a way to print it on one of the printers attached to your system. The PRINT command is described in detail in the next chapter.

Moving Data to a Disk File

Quite often you want to move data from the screen to a disk file. This can be invaluable when creating program documentation, for example, where you must create a textual representation of the screen display.

You can create this disk file, of course, by using the CUT command to select the screen (or desired portion thereof) and then using AlphaVUE (or similar program) to create a file into which you PASTE the text. While this is the most flexible method, it can sometimes be simpler to directly create an ASCII text file containing the selected portion of the screen.

MULTI provides such a capability by allowing an optional argument on the CUT command. By following the CUT command with a filename, you can specify a file to copy the selected text to.

For example, to store the selected text in the file SCREEN.TXT, type the command **`CUT SCREEN.TXT`**RETURN. You are then prompted to select a portion of the screen, as with any other use of the CUT command. Upon completing your specification of the portion of the screen to be copied, MULTI not only transfers it into the scrap area as usual, but also places a copy into the specified file.

Copying Data From a Disk File to a Window

Just as it is sometimes handy to be able to store a selected portion of a screen directly into a disk file, it can also be handy to be able to directly copy the contents of an ASCII disk file into a window. While it is possible to display the disk file, cut it, then paste it, you can save several steps by directly copying the disk file into the window.

MULTI provides this capability via an optional argument on the PASTE command. By following the PASTE command with a filename, you can specify an ASCII text file as the source of the data to be pasted, rather than using the scrap area as the data source as would normally happen.

For example, to copy the contents of the disk file DATA.TXT into a window, position the cursor where you want the data inserted, call up the MULTI Menu, and type **`PASTE DATA.TXT`**RETURN. The contents of the disk file are now inserted into the window.

If you cut data into a file, and don't specify a disk or account, MULTI places the file into the account you were logged into when you brought up MULTI, not the account the current window is logged into. If you paste data from a file and don't specify a disk or account, MULTI assumes the file is located in the account you were logged into when you called up MULTI.

CHAPTER 8

MULTI'S DESK ACCESSORIES

Introduction

This chapter discusses the special *desk accessories* available to you when running MULTI. In this chapter you will learn about:

- Using the desk calculator
- Using the perpetual calendar and calendar notecards
- Using the notepad
- Using the alarm clock
- Using the phone list
- Using the printer functions
- Using the directory command

MULTI stores information about the desk accessories—such as your phone list and notepad contents—in disk files. These files are generally created in the account and disk you are logged into when you first start MULTI. If you are the only person using MULTI in that account then these files become "your" phone list, notepad, etc. If you share that account with other users, you will share desk accessory settings. Therefore it is important you set up your own account in which you call up MULTI, and that you not share that account with others.

For information on the exact names and locations of the files MULTI creates, see Appendix A.

Using the Desk Calculator

Because it is one of the most frequently used tools on anyone's desk, MULTI provides a 10-digit, four-function calculator. This calculator is available at all times, regardless of what program you are running. In addition, you can cut and paste results from the calculator into other windows.

Calling Up the Calculator

To call up MULTI's calculator, call up the MULTI Menu. Now either type **CALCULATOR** RETURN or **CA** RETURN, or press SHIFT/F1. The calculator display now appears on your screen.

Each time you call up the calculator, it is in the same condition in which you left it last time you used it. This is handy when you are making a series of calculations and must go from your application program to the calculator and back again several times. Of course, if you exit from MULTI the calculator is cleared.

The calculator display represents a simple calculator to remind you of the functions that are present and the keys you use to invoke them. As you become more proficient in the use of the calculator, you may wish to eliminate the depiction of the calculator keys and have only the calculator's digit display visible. You can do this by typing **CALCULATOR SMALL** RETURN to invoke the calculator from the MULTI Menu. All functions work as before; this mode simply leaves more of the underlying screen display visible. If you wish to have the calculator always displayed in this small mode, you can place a SMALLCALC command in your MULTI.INI file, as discussed in Chapter 6. If you choose to place a SMALLCALC command in your MULTI.INI file, you can override it when calling up the calculator by typing **CALCULATOR BIG** RETURN.

When used on an AM-62, AM-62A, AM-72, or compatible terminal, the function keys immediately surrounding the numeric keypad are assigned to entering calculator commands, rather than their normal text editing functions. The assigned functions are displayed by their relative position within the calculator display on your terminal screen. This allows ten-key users to keep their hand in one position and still be able to access all functions of the calculator.

Users of other terminals will find all calculator functions available but assigned to other keys, as described below.

Entering Numbers

Enter numbers into the calculator by pressing any of the numeric keys on your keyboard. You may use either the numbers across the top of the keyboard, or the ones contained in the ten-key pad, if your terminal is so equipped.

In addition to the digits themselves, you can also use the [-] key to enter negative numbers, and the [.] key to enter a decimal point.

The Arithmetic Operators

The calculator supports the four basic arithmetic operators: addition (+), subtraction (-), multiplication (*), and division (/). Use these keys to enter the operation you wish to perform.

For the convenience of experienced ten-key operators, the [ENTER] key duplicates the function of the [+] key, as does the [RETURN] key.

The result of each calculation displays whenever you press the next operator, and calculations are done in the order in which you enter the numbers and operators. To terminate a series of operations, press the [=] key.

Alpha Micro terminal users can use NEXT WORD for multiplication, PREV WORD for division, and NEXT SCREEN for the = function.

The Memory Functions

MULTI's calculator supports a simple memory function. To store the number currently in the calculator's display in the memory, press [S].

After storing a number in the memory, you can call that number back into the calculator at any time by pressing [R].

On Alpha Micro terminals you can use INS CHAR to store a number and INS WORD to recall it.

The CUT Function

The calculator allows you to "cut" the value currently displayed into MULTI's scrap area, allowing you to later use MULTI's PASTE command to enter that number into your application software or files.

To place the currently displayed number into the scrap area, press [T]. Alpha Micro terminal users can also use INS LINE for the cut function.

See the discussion of the CUT and PASTE functions in Chapter 7 for more information.

Clear and Clear Entry

To clear the calculator display, press [C]; to clear just the last entry, type **CE**.

On Alpha Micro terminals, you can use PREV SCREEN to clear the display, and the comma key to clear the last entry.

Leaving the Calculator

When you are done using MULTI's calculator, press [MENU] to return to the MULTI Menu.

Alpha Micro terminal users can press [CANCEL] to leave the calculator (turn it off).

Using the Perpetual Calendar

MULTI supplies your terminal with a desktop perpetual calendar. You can display this calendar at any time by entering the MULTI Menu and either typing **CALENDAR**[RETURN] or **CALE**[RETURN] or by pressing [SHIFT]/[F2].

MULTI then displays the calendar page appropriate for the current system date. Or, if you have previously used the calendar during the current MULTI session, you see the month you last looked at. Note that today's date is highlighted if the current month is shown.

Turning the Calendar Pages

To turn the pages of the calendar, MULTI supplies you with several options. To move forward a month, press [↓]; to move back a month, press [↑]. You may use these two keys as many times as you like to select the page you wish to view.

NEXT SCREEN moves you forward one year each time you press it. PREV SCREEN moves you backward one year each time you press it.

To return to the calendar page showing the current date, press [HOME].

Using Calendar Notecards

The calendar desk accessory also provides a daily notecard facility, providing a simple place to put to-do lists, appointments, reminders, etc.

When you call up the calendar, by using [←] and [→], you can position yourself to any date of interest on the calendar page, highlighting that date. Pressing [RETURN] at this point calls up the notecard associated with that date. The manner in which dates are highlighted depends on your particular terminal—highlighted dates may be in color, may blink, etc.

The notecard works just like the notepad desk accessory except it is tied to the specific date you selected. You can record and edit whatever text information you like within the notecard. Pressing [MENU] returns you to the calendar.

The file DAYSCH.DAT is automatically loaded into the calendar notecard the first time it is called up. This sets up a default template for scheduling or other note taking. You can modify this file so all MULTI users can share it, or place different versions of the file in each user's login or root account. The search precedence for DAYSCH.DAT is: account MULTI user was logged into before bringing up MULTI, user's root account, and MULTI's library, MULTI:.

Leaving the Calendar

Once you are done examining the calendar, press [MENU] to return to the MULTI Menu.

Using the Notepad

The notepad accessory provides a convenient place for you to jot down notes on a phone call, maintain a "to-do" list, or replace any other of the many scraps of note paper that seem to cover the average desk. You can display this notepad at any time by entering the MULTI Menu and either typing **NOTEPAD**[RETURN] or **N**[RETURN], or by pressing [SHIFT]/[F4].

Entering and Editing Text

Type text on the notepad using the standard AlphaVUE editing commands. AlphaWRITE users will find that the most frequently used AlphaWRITE editing commands are available. The available commands are:

Arrow Keys	Move cursor in all four directions
[HOME]	Position cursor at top of notepad
[SHIFT]/[HOME]	Position cursor at end of notepad
[SHIFT]/[←]	Position cursor at beginning of line
[SHIFT]/[→]	Position cursor at end of line
NEXT WORD	Position cursor on next word
PREV WORD	Position cursor on previous word
NEXT SCREEN	Position cursor on next screen of text
PREV SCREEN	Position cursor on previous screen of text
INS CHAR	Insert a character space
DEL CHAR	Delete a character
DEL WORD	Delete a word (up to next space)
INS LINE	Insert a blank line
DEL LINE	Delete a line
[F16]	Toggle character insert mode
[TAB]	Indent text
[CTRL]/[Y]	Erase to end of line
[CTRL]/[O]	Concatenate lines

The notepad automatically wraps text as you near the end of a line. It will always break a line at either a space or tab. If you wish to disable automatic word wrapping, you can place a WORDWRAP=FALSE command in your MULTI.INI initialization file, as discussed in Chapter 6.

Specifying the Notepad Name

When you type **NOTEPAD** RETURN at the MULTI Menu or press SHIFT/F4, MULTI displays your default notepad, which it gives the name "NOTEPD.TXT." However, MULTI allows you to keep as many different notepads as you like. To specify a notepad other than the default, type a notepad name after typing the NOTEPAD command. For example, type **NOTEPAD TODO** RETURN to use the notepad named TODO to keep your "to-do" list.

Notepad names can be any six alphabetic or numeric characters. MULTI doesn't pay any attention to upper or lower case in notepad names, treating them the same.

Leaving the Notepad

When you are done updating the information in your notepad, press MENU to return to the MULTI Menu. The information you have typed into your notepad is then stored away until you call it back again later.

Using the Alarm Clock

MULTI supplies your terminal with an alarm clock you can use to remind yourself of upcoming meetings and other events. Unlike regular alarm clocks, you can set up to ten alarms to go off at different times, even months in the future! Included with the date and time that the alarm is to go off, short explanatory text reminds you of why the alarm is sounding.

You can call up the alarm clock accessory at any time by entering the MULTI Menu and either typing **ALARM** RETURN or **A** RETURN, or by pressing SHIFT/F3. MULTI then displays the current alarm clock settings.

When an alarm setting goes off, your terminal "beeps" and the text associated with the alarm displays in a window on your terminal screen. After a brief delay, you are asked to press any key to remove the alarm display so you can continue with your work.

MULTI tries to avoid interrupting your active work by an alarm going off, so it waits until you have stopped typing for a few moments before notifying you of an alarm. Of course, MULTI can only set off an alarm while you are working within MULTI.

Displaying the Current Settings

Whenever you call up MULTI's alarm clock, it displays the current alarm clock items that are set. The very first time you call this accessory up, of course, this display is empty. As you use it throughout the day, this display serves both to remind you of upcoming events, and to allow you to change items you have scheduled.

The list of items displayed within the alarm clock is ordered so the next alarm to go off is at the top.

Adding a Setting

Once you have selected the alarm clock accessory, add a new alarm clock setting by pressing the [A] key. A new item appears at the top of the item list, showing today's date and a time of 12:00 PM.

Select the date on which you wish this alarm item to sound. Enter this date by typing the first two digits of the date, pressing [RETURN] to advance to the next field, typing the next two digits, pressing [RETURN], typing the final two digits, and pressing [RETURN] one more time. If you wish to leave part or all of a date as it is shown, press [RETURN] to leave the field unchanged.

The cursor is now positioned on the hours field. Type two digits for the hours and press [RETURN]. Type two more digits for the minutes and press [RETURN].

Enter the time as your system is set up for: either 12-hour (AM/PM) or 24-hour format. If using AM/PM format, press [A] for AM or a [P] for PM and press [RETURN]. As with the date, if you wish to leave any one of these fields, use [RETURN] to step over the field.

With the cursor now positioned on the "Alarm" field, type the message you wish to see displayed when this alarm sounds. Press [RETURN] when you are done.

You may now press [RETURN] to step through the various fields, updating them if you changed your mind about the date, time or message. When you are satisfied with the setting, press [MENU]. The new item is now inserted in the list of settings, which is ordered by when the alarm will go off.

Deleting a Setting

To delete an alarm setting, press the [D] key. A large cursor appears highlighting the first entry. Now use the [↑] and [↓] keys to select one of the alarm clock entries with the large cursor that has appeared. Once this cursor is positioned on the item you wish to delete, press [RETURN]. The item is now deleted from the alarm clock listing.

Note that alarm clock items are automatically deleted from the alarm listing once they have gone off.

Changing a Setting

To change an existing alarm clock setting, press the [C] key. A large cursor appears highlighting the first entry. Now use the [↑] and [↓] keys to select one of the alarm clock entries with the large cursor that has appeared. Once this cursor is positioned on the item you wish to change, press [RETURN]. This item is now selected for change. Press [RETURN] to step through the various fields, changing them as you wish.

When you are satisfied with your changes, press [MENU]. The newly updated item is now inserted back into the alarm settings list in its proper order.

Leaving the Alarm Clock

When you are done updating the alarm clock settings, press [MENU] to return to the MULTI Menu.

Using the Phone List

MULTI's phone list not only stores all the phone numbers and addresses you need, it lets you search the list according to any part of the information you've entered, so you can see only those people with a certain last name, in a specific area code, who work for a specific company, and so on. You can also keep a log of the calls you make, and if you have an auto-dialer attached to your system, have MULTI dial your calls for you.

You can display the phone list at any time by entering the MULTI Menu and typing **`PHONE`**[RETURN] or **`P`**[RETURN] or **`PHONE phone-list`**[RETURN] (where **`phone-list`** is a specific phone list file), or by pressing [SHIFT]/[F5]. If you do not specify a phone list name, the PHONE command automatically calls up the phone list PHNLST.DAT, or creates it if you do not yet have a phone list. The phone list file is created in the disk account you are in when you bring up MULTI, and maintained there when you make changes to it.

By specifying different phone list names (six characters or less) to the PHONE command, you can create different phone lists for different purposes (for example, personal phone numbers, business leads, vendors, etc.). Phone list files always have the extension .DAT.

When you give the name of a phone list file to MULTI, make sure the file really is a MULTI phone list, or unpredictable things will happen!

If you prefer to maintain a single phone list, use the PHONE command without specifying a phone list name.

The first time you call up the phone list, MULTI asks you how big to make the phone list. MULTI needs to reserve space to store the phone list information. If MULTI asks you to specify how big the phone list should be, type the number of entries you will want to store and press [RETURN]. Make sure you specify enough entries to store all of the phone numbers you will want to use. If you need to expand the phone list later, you can do so—see later in this chapter for details.

After using the PHONE command, or once MULTI has allocated space for the phone list, MULTI shows a window giving you a summary of the items you have stored in your phone list. This list displays in alphabetical order, based on last name. If your list is too long to be displayed in the window, you can move forward and backward in the list via the NEXT SCREEN and PREV SCREEN keys, just as when you are using the notepad accessory. You can also return to the start of your list by pressing the [HOME] key, or move to the very end of the list by pressing [SHIFT]/[HOME].

This summary display is probably the way you will most commonly use the phone list accessory, as it shows you the most commonly desired information for each phone list entry.

Adding a New Item

To add a new phone list item, press the [A] key while in the summary listing. You next see a window showing a blank entry form. Type information into any of the fields you find applicable, pressing [RETURN] to complete each field and move on to the next.

You can move around through this form by either pressing [RETURN] or the [↑] and [↓] keys. The only field which must be filled in is the "Last Name" field; all others can be used or not, whatever you find useful.

One of the fields in the entry form asks if you want to maintain a phone log for this item. A phone log is a text file that provides a permanent record of any calls to this person. There are some restrictions on using phone logs, so see "Placing and Logging a Phone Call" later in this chapter before answering "Y" to this field.

Once you are satisfied with the information you have added, press [MENU] to return to the summary listing window. Your newly added item is inserted into the summary listing in alphabetical order.

Displaying the Individual Phone List Items

In addition to the information displayed within the summary window, the phone list accessory keeps additional information on each entry. To display this more detailed information, select a phone list item by using the [↑] and [↓] keys to move the highlighting cursor over the entry you wish to display. Once you have selected an item with this cursor, press [RETURN]. The summary display window is replaced with one displaying detailed information about the selected entry.

Once you are done with this detailed display and wish to return to the summary listing, press [MENU].

Changing an Item

To change an item within the phone list, first select the item for detailed display as described above. Once the detailed display window is visible, you can move from field to field pressing [RETURN] and the [↑] and [↓] keys.

After placing the cursor on the field you wish to change type the new data. You may change as many different fields as you wish.

Once you have made all of your desired changes, press [MENU]. You then return to the summary listing.

Deleting an Item

To remove an item from the phone list, position the highlighting cursor over the item you wish to delete, just as when selecting the detailed display. Once you have selected the item, press the [D] key. The item is now deleted and is removed from the display.

You will not be asked to confirm you want to delete the item, so make sure you have selected the proper item before pressing the [D] key.

Selecting Specific Items

As your phone list grows, it becomes harder and harder to find a specific number if you don't remember the person's exact name. MULTI helps you with this problem by letting you select phone numbers according to any of the information you've entered for them. For instance, you can view only those people in a certain state or area code, who work for a particular company, and so on.

To do this, press [S] to enter selection mode. You see a blank phone entry form. Fill in the field(s) you want to use to make your selection, leaving others blank. For example, if you want to see everyone in California who makes widgits, type "widgit" in the keywords field and CA in the state field, leaving the rest of the form blank. For all fields but the Name field, you can enter partial information (for example, just a "W") to select items whose field

contains the characters you entered. For example, a keyword of "W" selects "widget," but also selects "Awnings" and "waxworks." When you're done entering your criteria, press MENU.

The phone summary list redisplays, but it now includes only the items meeting *all* of the criteria you entered. In our example, you see only those people who both live in California and have a keyword containing the letter "W."

If after selecting the list of numbers is still too big, you can use the Select command again to further qualify the list. Each time you use the Select command, you further qualify and reduce the list. If you specify criteria that prevent any items from being selected, you end up with an empty list.

To return to a full summary list, press S and then press MENU without filling in any fields.

MULTI always shows the number of selected fields in the upper right corner of the phone list accessory.

Placing and Logging a Phone Call

If your system has an automatic dialer attached, MULTI can automatically dial a phone number for you. Select the number you wish to have dialed by moving the large cursor over the number to be dialed and pressing C for "call."

The automatic dialer then dials the phone for you. After dialing, MULTI prompts you to pick up the phone.

If you have enabled phone logging for the number you are dialing, MULTI then calls up the appropriate phone log, adds the current time and date to the log, and allows you to enter text into the phone log and/or review the notes you took on previous calls.

All of the same editing commands which are available in the notepad accessory are also available in the phone log.

Once you are done working with the phone log, press MENU to return to the phone list accessory.

Phone logs can be kept even if you don't have an automatic dialer by dialing the number manually, and then pressing C to make a phone log entry.

By using the SETUP command, you can select whether a single phone log will be kept for all phone numbers, or a separate log for each phone number in your phone list will be maintained. See Appendix A for details.

Using phone logging for more than one phone list in a single account can cause phone logs to be scrambled.

Using phone logging is optional. If you want to use phone logs, choose a single phone list in your MULTI account with which to use phone logging; make sure all items in other phone lists in that account have an "N" next to the "Maintain phone logging" field in their entry forms.

Looking at a Phone Log

There are times when you must review your log of phone calls without actually placing a call or making a new entry in the phone log. You can do this by selecting the proper phone number with the arrow keys and pressing [L] to view the phone log.

The phone log then displays on your screen and you are free to examine and change it as you wish. When you are done looking at it, press [MENU] to return to the phone list accessory.

Leaving the Phone List

When you are done using the phone list accessory, press [MENU] to return to the MULTI Menu.

Printing the Phone List

MULTI can print the phone list for you in two different formats. For information on how to print the phone list, see the description in "Using the Printer Function," below.

Expanding the Phone List

If you run out of room in the phone list, you will need to expand it. The SETUP command allows you to expand an existing phone list to make room for new entries. To expand a phone list, at the MULTI Menu type **SETUP**[RETURN] or **SET**[RETURN], or press [F8]. Within the SETUP Menu, press [F4]. You are asked for the name of the phone list you want to expand. If you want to expand the default phone list PHNLST.DAT, press [RETURN]. Otherwise, enter the phone list filename and press [RETURN]. Next you are asked for the number of empty entries you want to add. Enter this number and press [RETURN]. After a brief pause, the phone list is expanded and you return to the SETUP menu. To exit the SETUP menu, press [MENU].

When you expand a phone list, make sure you give MULTI the name of an actual MULTI phone list file. Expanding a file that is not a phone list causes problems.

Using the Printer Function

As another one of its desk accessories, MULTI provides flexible access to the system printers for a variety of purposes. You can print a copy of the current scrap area (which you have put data into via the CUT command), you can print a copy of the current screen image, and you can print a disk file.

In addition to allowing you to print on any of your system printers (which are controlled by the system printer spooler), MULTI also allows you to print on a local printer attached to your terminal, if you have one.

To invoke the print accessory, call up the MULTI Menu and either press [SHIFT]/[F6], or type **PRINT**[RETURN] or **PR**[RETURN].

The print accessory then shows a command window asking what you wish to print. Make your choice and press the appropriate function key.

Printing the Current Scrap

If you press [F1] to select printing of the current scrap area, you must have previously cut some text using the CUT command. If you have not, MULTI displays an error message and asks again what it is you wish to print. Press [MENU] to return to the MULTI Menu and cut some text before using the print command again.

If you have previously cut text, MULTI prompts you for a choice of printer. If you have previously used the SETUP command to tell MULTI that you have a local printer attached to your terminal, the local printer will be one of your choices. Select the printer you wish to use and press [RETURN]. If you only have one printer on your system, this step will be bypassed and printing starts immediately.

Printing a Screen Image

By pressing [F2] you can select the printing of an image of the screen currently displayed on the current window whose name appears at the bottom of the screen. This option can be useful when preparing documentation for a software package or to take a quick "snapshot" of some important results displayed on the screen.

After selecting the print screen option, MULTI prompts you for a choice of printer. If you have told MULTI (via the SETUP command) that you have a local printer attached to your terminal, the local printer will be one of your choices. Select the printer you wish to use and press [RETURN]. If you only have one printer on your system, this step is bypassed and printing starts immediately.

Printing a File

Although most applications provide a way of sending files to the printer, there are still times when it is handy to be able to send a file to the printer without stopping what you are doing.

After pressing [F3] to select printing of a file, MULTI prompts you for the name of the file you wish to print. Enter any standard AMOS file specification, and press [RETURN].

MULTI now prompts you to select the printer to use. If you have told MULTI (via the SETUP command) that you have a local printer attached to your terminal, the local printer will be one of your choices. Select the printer you wish to use and press [RETURN]. If you only have one printer on your system, this step is bypassed and printing starts immediately.

Printing a Phone List

The MULTI PRINT command allows you to print the phone list you have created with the phone list desk accessory. Call up the PRINT command and press [F4].

MULTI requests the name of the phone list you wish to print. If you are printing the default phone list, PHNLST.DAT, press [RETURN]; otherwise, type the name of the phone list you wish to print and press [RETURN].

Next MULTI requests the format you wish the phone list printed in. The first selection, [F1], produces a brief listing with a single line for each phone list entry. The second selection, [F2], prints a full listing of all information contained in the phone list.

Finally, MULTI asks for the printer you wish to print the phone list on. Select the desired printer by typing the appropriate number.

After a brief pause for formatting the phone list, MULTI begins printing.

Maintaining the Printer Queue

MULTI provides a convenient way to examine and change the files waiting in the printer queue via its printer queue maintenance option. By selecting [F5] within MULTI's PRINT command, you can:

- Display the list of files currently in the printer queue
- Remove a print request for a file from the printer queue
- Change the priority of a print request

- Specify the name of the form currently mounted on the printer

Displaying the Printer Queue Maintenance Screen

To call up MULTI's printer queue maintenance screen, call up the PRINT command from the MULTI Menu. When asked for which printer function you wish to perform, press [F5] for Maintain Printer Queue. If you have more than one printer defined, you are then asked to select the printer for which you wish to see the queue. If only one printer is defined, you see the printer queue maintenance screen immediately.

After selecting a printer, if you needed to, you see a window displaying the current contents of the selected printer's queue, along with three options: REMOVE, MOUNT, and SELECT. You can choose any one of these three options, or use the queue maintenance command simply to look at the contents of the queue without changing anything.

If all you needed to know is what files are currently in the queue, or just get an idea of how long a file is going to take to print, this screen gives you that information at a glance. When you are through, press [MENU] without selecting a queue maintenance option and you return to the PRINT option screen.

If there are more files in the queue than will fit in the queue maintenance window, use the [↑] and [↓] keys to display the remaining files. The [↓] key moves the display forward through the queue list and the [↑] key moves the display back through the queue list.

If there are no print requests in the printer queue, you see the message "The printer queue is empty."

Removing Documents From the Printer Queue

Occasionally you may request printing of a file and then change your mind before printing actually begins. The REMOVE option lets you remove a document from the printer queue and "cancel" the print request. The file itself is not changed and, if you choose, you can request printing at a later time.

The REMOVE option also allows you to cancel a print request after the file has started printing. Your printer might have a paper jam, the ribbon or paper might have run out, or you may need to stop the printer from printing

the document—if this is the case, you can use the REMOVE option to stop the print currently in progress.

To remove a print request from the printer queue or stop a print that is in progress, type **R** RETURN in response to the prompt "Enter printer option:".

A new prompt then appears: "Enter number of selected file:." After viewing the printer queue, determine the number of the file you wish to remove from the printer queue. Type the file number you have selected and press RETURN.

The printer queue display then changes to show the selected file is no longer waiting to be printed.

You can only remove files from the printer queue that are in the same disk account project (same [p,]) as MULTI was originally called from. This is to prevent you from accidentally removing other users' files by mistake.

If you have removed a print request for the file currently being printed, it may take a short time before the printer actually stops.

Specifying a New Form on the Printer

A new form name must be specified when the form name associated with the file at the top of the printer queue is different from the "Current Form" name displayed on the menu screen. After installing the new form on the printer, the MOUNT command can be used to notify the computer that the new form has been installed and that printing may resume.

To instruct the computer that the new form has been installed on the printer, type **M** RETURN. You then receive the prompt "Enter name of new form:".

Type the new form name, of six characters or less, associated with the file and press RETURN. The current form name on the top line of the queue display is then updated to reflect the new form name you specified. All remaining documents with this new form are printed—no matter what their priority in the printer queue. If another file with a different form name remains in the printer queue, change the form on the printer and use the MOUNT option again to specify the new form name and continue printing.

Changing the Priority of Print Requests

The printer queue normally functions as a waiting list of files to be printed, organized on a first come, first served basis. There may be times when this is inconvenient—a lengthy document may be queued up and there is a rush print request to be processed. When this occurs, you can use the SELECT option to change the order (or priority) of the print requests.

As a matter of fact, through repetitive use of the SELECT option, you can cause the printer queue to be arranged in practically any order you want.

The SELECT option, unlike the REMOVE option, allows you to change the priority of any file in the printer queue, regardless of the project number it comes from. The only exception is the first file in the queue, which is already being printed, and whose priority cannot be changed.

To change the priority of a print request, type **S** RETURN in response to the prompt "Enter printer option:". You are then asked to "Enter number of selected file:."

Determine the number of the file that should be placed as the next file in the printer queue. After typing the file number and pressing RETURN, the queue display updates to show the selected document is in line to be printed next and is the second document. Documents following in the printer queue then move down in priority and on the printer queue list.

Using the Directory Accessory

In many programs, you often need to type the name of an existing disk file. While usually this can be quickly provided from memory, there are times when it becomes hard to remember the exact name of the file you want—"Did I call that report FILE1, FILONE, FIL1 or FILE01?" MULTI's directory accessory can help you find the file you want.

The directory accessory lets you get a listing of the files in a disk directory without leaving the current application program. Along with the alphabetized list of file names, MULTI also displays a brief description of the file. Wildcards (discussed below) allow you to select only those files you wish to see.

Displaying the Current Directory

Call up the directory accessory by either pressing [SHIFT]/[F7], or by typing **DIR**[RETURN] or **D**[RETURN]. MULTI then opens up a display window showing an alphabetized list of files and their descriptions currently contained in the directory where the current window is logged in.

If more than one page of files is present, you can press NEXT SCREEN and PREV SCREEN to page forward and backward within the directory display. You can go to the beginning of the directory by pressing [HOME]. You can also abort the directory display and return to the MULTI Menu at any time by pressing [MENU] or [SHIFT]/[MENU].

Displaying Other Directories

You can display the contents of directories other than the one the current window is logged into by specifying either a device or account or both on the command line with the DIR command. For example, to display the contents of the directory DSK1:[100,5], at the MULTI menu type:

```
DIR DSK1:[100,5]RETURN
```

Using Wildcard File Specifications

MULTI's directory command supports the use of both ***** and **?** wildcard symbols to allow you to display only selected files within a directory. For example, all of the following are valid MULTI directory commands:

```
DIR DSK1:*.WRT[100,5]RETURN
DIR AB???.M68RETURN
DIR *.BASRETURN
DIR DSK2:*.00?RETURN
```

The directory command does not support wildcarding within account specifications, only within file names and extensions. See the *AMOS User's Guide* for more information on wildcards.

CHAPTER 9

CHANGING WINDOW CONFIGURATIONS

Introduction

This chapter discusses how you can change a window's parameters once you have opened it. You will learn about:

- Changing a window's default application
- Changing other window parameters

Changing Window Parameters

The CHANGE command allows you to examine and change all window parameters, including default memory allocation, and default application.

Select the CHANGE command by pressing [F6] or by typing **CHANGE** [RETURN] or **CH** [RETURN].

A display of options appears within the CHANGE command window. Use the [↑] and [↓] keys to move the cursor between the various fields shown, which are described below. Once the cursor is positioned on the field you wish to change, enter the new data and press [RETURN]. Once you have changed the fields you want to, press [MENU] to return to the MULTI Menu.

After performing a window name change, the screen is re-displayed, showing your changes. Other fields, such as memory size or startup application have no immediate effect on the window. All you have changed is the information that will be used to open the window the next time MULTI is invoked from AMOS command level—and remember, even that information will be lost if you do not save a window configuration file, via the SETUP command's Save option, before exiting MULTI.

The Window Name

The CHANGE command allows you to alter the name assigned to the window. You can use any name of up to 15 characters, including spaces. You can use upper or lower case, and while MULTI displays the name in upper and lower case, it ignores the case when comparing names for duplicates or when selecting a window.

To make the name change permanent, you must use the SETUP command to save the new configuration.

Don't use a MULTI Menu command for a window name (e.g., PHONE), or you won't be able to select that window at the MULTI Menu by just typing its name. Instead, you will have to use one of the alternate methods of selecting a window.

Memory Partition Size

CHANGE allows you to alter the amount of memory which will be assigned to the window when it is opened. You can assign any amount you wish, as long as it is 2048 bytes or more. You can include the multipliers K or M when specifying a memory size.

Changing the memory size with the CHANGE command has no immediate effect. It only alters the memory partition size of the window if the window configuration is saved and MULTI is restarted.

Initial Command Line

The "initial command line" is used to start a window when MULTI first opens it during startup, and specifies what the window will do. You can use the CHANGE command to alter this command line and thereby change what the window starts running when MULTI is started.

You can specify any command line you wish, up to 60 characters in length. This command line is sent to the window when it is first started and is at AMOS command level.

Changing the initial command line has no immediate effect on the window, and only takes effect if you save the current window configuration and restart MULTI.

APPENDIX A

INSTALLATION AND CONFIGURATION REQUIREMENTS

Introduction

This appendix describes the system configuration required to use MULTI, and how to install it on your computer. It also describes some of the differences you can expect to see in various hardware environments.

Minimum Requirements

- Minimum user memory partition of 200K bytes when a memory pool is not set up
- CRT terminal with direct cursor addressing, clear to end of line and clear to end of screen

Memory Requirements

MULTI requires at least a 200K byte user partition to be used effectively. Since each window you open requires a memory partition (which you specify the size of) plus approximately 18K of overhead, the number of windows you may open is directly proportional to the amount of memory you have available in your partition. A typical user with 3-4 windows open can easily use up a 500K memory partition. This makes sense when you consider that three open windows are functionally the same as three separate people running separate tasks and thus require about the same amount of memory.

MULTI alternatively allows you to define a central memory pool from which all MULTI users can draw, letting you decrease the amount of memory required by each individual MULTI user. See "Allocating Memory for MULTI," later in this appendix for details.

AMOS Version Prerequisites

MULTI makes extensive use of the latest AMOS features. Please check the Release Notes for this version of MULTI determine the AMOS version requirements.

MULTI is optimized to work under the AMOS 2.x system. MULTI will work under the older 1.x systems, but with reduced performance.

Where MULTI Creates Files

As you make use of the MULTI desk accessories, MULTI creates and maintains several disk files. Here is what they are called and the functions they serve:

FILE	FUNCTION
TICKLR.DAT	Alarm clock settings.
yymmdd.DAY	Calendar notecard. The name of the file represents the calendar date you created the notecard for. The first two digits represent the year, the second two digits represent the month, and the last two digits represent the day. For example, 890425.DAY is created for April 25, 1989.
nnnnnn.PLG	Phone log. If you have enabled a single phone log for all your phone calls, this file is named PHNLOG.PLG. If you have enabled multiple phone logs, multiple phone log files will exist, each with a unique numeric name (e.g., 000024.PLG).
xxxxxx.xxx	Notepad file. When you create a notepad, you may supply any six-character name. If you don't supply a notepad file name, MULTI uses the default name NOTEPD.TXT. If you don't supply an extension, MULTI uses the default extension .TXT.
xxxxxx.DAT	Phone list. When you create a phone list, you may supply any six-character file name you want. If you choose to create a phone list and do not supply a name, this file is created with the default name PHNLST.DAT.

Each of these files is created and maintained in the account the MULTI user logs into before calling up MULTI, except for TICKLR.DAT, which is created and maintained in the user's root account. The root account is defined and set by the AMOS MUSER command. If your computer does not use the MUSER user name feature, TICKLR.DAT is created in the account the user is logged into before calling up MULTI.

Installing MULTI on Your System

Installing MULTI on your computer is a straightforward operation, but usually requires you to reboot your computer at least once during the installation process. For this reason, it is best to schedule the installation for a time when the other users of your computer will not be affected by the installation process.

Installing MULTI's Files

The first step in installing MULTI is to copy the files which make up MULTI onto your system disk using the method appropriate for the media on which you received the MULTI software. You should copy all of the files present on your release media to your system disk. They are stored on the distribution media in the accounts they should be stored in on your disk.

If you are installing MULTI from the AlphaCD Software Distribution CD-ROM, then installing MULTI is simply a matter of copying the files from the appropriate logical unit on the CD-ROM.

If you are installing MULTI from VCR tape, type:

```
LOG OPR: RETURN
VCRRES DSK0:[] = ALL:[] RETURN
```

No matter how you are installing the software, MULTI places most of its files, including overlays and help files, into account DSK0:[7,15]. If your computer does not have enough space on DSK0: to accommodate these files, MULTI allows you to place them elsewhere on your computer. When searching for its files, MULTI first checks in the ersatz device MULTI:. Only if no MULTI: device exists is DSK0:[7,15] searched. By defining a MULTI: device that references another account and device, you can place MULTI's files anywhere on the computer.

Included with the MULTI software on the distribution media is the file MULTI.DIR. This file contains a complete list of all of the files making up the MULTI system, along with the hash total of each file. You can use this listing, and the VERIFY program, to make sure all of MULTI has been properly installed on your computer. To verify the MULTI files have properly been copied, type:

```
LOG OPR: RETURN
VERIFY MULTI RETURN
```

Creating WINDOW Translation Files

In order for function keys to work for your terminal when you are using windows, you must create the WINDOW function key translation files. To do this, type:

```
LOG OPR: RETURN
COPY WINDOW.* = DSK0:AM62A.??X[7,*] RETURN
```

Since all Alpha Micro terminals in the AM-6X, AM-72, and AM-75 family use essentially the same function key translation, this creates function key translation files that can be used by all Alpha Micro terminals when using MULTI windows.

For information on how to enable function key translation for non-Alpha Micro terminals, see "Setting Up for a Non-Alpha Micro Terminal," below.

Modifying the System Initialization File

After copying the MULTI files to your system disk, log into DSK0:[1,4] so you can modify your system initialization file. As usual, never make changes directly to the system initialization file, but instead make a copy and modify that copy. Only once the system has been successfully booted using this modified copy should you replace your current initialization file with the modified one.

Edit your copy of the initialization file and locate the line (near the top of the file) which contains the "JOBS" statement. The JOBS statement includes a number, which tells the computer how many job "slots" to create during system boot-up. Every job on the computer must have a "slot" assigned to it. Typically, the number following the JOBS statement reflects the number of jobs allocated on the computer. You must increase this number to allocate additional job "slots" for MULTI to use when opening windows for you.

As a minimum, you must allocate one additional job slot for each window which any user of the computer might have open. Each additional job slot requires only four bytes of memory, so it is wise to be generous when allocating the additional slots. Allocating ten additional job slots for each user which may use MULTI, just in case they want a lot of windows open, would not be excessive.

At this time you may also wish to change the memory allocated to the jobs on your computer. Such changes may be necessary to ensure that those users who will be using MULTI have sufficient memory assigned to them.

After modifying the copy of the initialization file, use the MONTST program to test your changes. The computer should boot normally. After the computer has completed booting, use the

JOBS command at AMOS command level to verify the correct number of jobs have been allocated. Use the SYSTAT command to make sure any changes in memory allocation you made have also worked correctly.

Once satisfied with the modified initialization file, you may rename the modified version as AMOSL.INI or AMOS32.INI, as appropriate, so it is used each time the computer is booted.

For further information on system initialization files, refer to the *AMOS Operator's Guide to the System Initialization Command File*.

Defining the Available Printers

MULTI first looks for a MULTI.INI file in the account you are logged into when you bring up MULTI, and sees if printers are defined in that file.

If no MULTI.INI file exists in which printers are defined, MULTI looks for the file PRTDEF.INI in DSK0:[1,4] for its printer definitions, just as AlphaWRITE and AlphaCALC do. If MULTI does not find a file called PRTDEF.INI, it looks for the file PRTDEF.lan in DSK0:[1,4], where "lan" is the language dependent extension defined in the language definition file your job is using.

If neither PRTDEF file is present on your computer, you can add printer definitions to your MULTI.INI file, but it is probably better to create a PRTDEF file so the many different applications on your computer needing printer definitions share the same definitions in the single PRTDEF file.

Whether contained in PRTDEF.INI, PRTDEF.lan or MULTI.INI, the printer definition line appears the same:

```
PRINTER name,driver,"descr",wheel,feeder,cpu
```

where:

OPTION	DEFINES
name	The name of the printer as defined to the spooler.
driver	The name of the printer driver (.PDV file) used by this printer. The printer driver is not used by MULTI.
descr	A description of the printer to be displayed in menus, enclosed in quotes.
wheel	The print wheel number currently mounted on the printer.
feeder	A Y or N, depending on if you have a sheet feeder.
cpu	The CPU number of the computer where the printer is located if a remote printer is being defined. This may be a decimal number or an ersatz device.

For example, the following are all valid PRINTER statements:

```
PRINTER LPT0,MISC,"The Fast Draft Printer"
PRINTER DAISY,AM302,"Printer #2",2,N,16842753
PRINTER SMALL,AM306,"The Color Printer",,,CPU1:
```

MULTI supports up to 18 printers.

Installing the Product Installation Code

Once you have successfully booted your computer under the modified system initialization file, you must complete the installation of MULTI by entering your Product Installation Code (PIC). This PIC keys MULTI to work with the particular Software Security Device (SSD) contained in your computer. You will be given a PIC by the supplier of the MULTI software. To obtain the correct PIC, you need to provide your Alpha Micro dealer with the serial number of your SSD. This serial number can be found on the System History label located on the central processor cabinet.

The particular PIC you receive from your dealer authorizes you for a specific number of simultaneous MULTI users (using up to 16 windows apiece), so be sure to decide on the maximum number of users you think will use MULTI at one time before calling your dealer. Special PICs are available for demonstration versions of MULTI allowing a single user with a maximum of two open windows.

To install the PIC, run MULTI for the first time by typing:

MULTI RETURN

You see a screen display requesting your PIC. Enter this combination of letters and numbers, being careful to double-check that all digits and letters are correct, in the proper case, and with the correct punctuation; then press RETURN.

If the PIC was the proper one for your particular computer, after a brief pause you return to AMOS command level. The installation of MULTI is complete. You may now call up MULTI again and use it.

If you receive an error message instead, check to be sure you have entered your PIC properly, and you have the correct PIC for the Software Security Device installed in your computer.

While entering the PIC, all users are locked out of the computer for just a few seconds until you finish entering the PIC.

If you have doubts as to whether the PIC was entered correctly, you can display the PIC code that MULTI is keyed for by typing the command:

MULTI/P RETURN

Each PIC authorizes a certain number of simultaneous MULTI users. For example, a PIC that authorizes four simultaneous users allows up to four people to run MULTI at the same time, regardless of how many windows they have open, or how many other users are on the computer running other programs.

If someone tries to use MULTI and the maximum number of simultaneous users are already using MULTI, they receive an error message. They must then wait for another user to exit MULTI before they can enter it.

Be sure you know how many simultaneous users your PIC authorizes you for.

Allocating Memory for MULTI

Normally MULTI allocates memory for its windows within the user's partition. This is the simplest method as it requires no special setup on the part of the system operator. If you are setting up MULTI for the first time, we recommend you start by leaving MULTI in this default memory allocation method until you are familiar with MULTI and can better estimate the memory requirements of the MULTI users on your computer.

One of the advantages to allocating memory within the user's partition is the repeatability of the memory allocation. Unlike a central memory pool, where multiple users all compete for the limited resource of memory, the user's partition is of a fixed size available only to that one user. What is available one day will be available the next, regardless of what other users on the computer are doing.

In some circumstances, however, this can be a disadvantage. If the users on your computer perform greatly varying tasks, needing only a single window at one time and five windows another, pre-allocating sufficient memory to each user to handle the "five window" case may result in wasted memory during the time when only one window is needed. For this reason, MULTI supports an optional central memory pool.

When MULTI gets a request for memory, it first checks the user partition, just as before, to see if space is available. If it is, it is allocated in the user's partition and the request is satisfied. If there is no room in the user's partition, MULTI scans for a central memory pool located in system memory and attempts to allocate memory within that pool. If there is memory available, the request is satisfied. Only if no memory is available in the central pool, or if no central pool was allocated, is an "insufficient memory" error returned.

This method seems to offer an ideal compromise: those who prefer the simplicity and predictability of allocating memory within the user's partition can leave things as they are, allocating additional memory to each MULTI user as required. Those who prefer a central memory pool can allocate one and reduce the user's partition size in proportion. By supporting both methods, MULTI provides the benefits of central allocation while still allowing effective use of whatever extra memory the user's partition may contain.

The central memory pool is allocated by the statement:

```
SYSTEM MLTMEM.SYS/N x
```

inserted into the rest of your SYSTEM commands in the AMOSL.INI or AMOS 32.INI file. The value ***x*** specifies the amount

of memory to allocate and can contain the multipliers K or M for kilobytes and megabytes. Thus a typical command line might be:

```
SYSTEM MLTMEM.SYS/N 500K
```

Reducing Memory Requirements

MULTI supports terminals with up to 50 lines on the screen. Because a program can change the number of displayed lines at any time, MULTI must allocate memory for up to 50 lines, regardless of whether you ever use it.

If you do not use terminals with more than 24 lines, you can save some memory (approximately 9.5K per open window) by adding a ROWLIMIT command to the MULTI.INI initialization file. The format of this command is:

```
ROWLIMIT=value
```

where ***value*** is the maximum number of screen rows you wish to be able to display. For example, if you never use more than 24 rows, you would add the command `ROWLIMIT=24` to your MULTI.INI file. The default row limit is 50, so any limit below that will save some memory.

Using the SETUP Command to Define Available Applications

The OPEN command provides a quick way of setting the startup application by providing a list of available applications. MULTI comes with a default list of applications, some of which may not be available on your computer. Or, the list may not include important applications you do have. For this reason, the SETUP command allows you to modify this list of available applications to match the applications you have installed on your particular computer. If you are installing MULTI for the first time, however, you may wish to skip over this process for now and come back later when you are more familiar with MULTI.

To maintain this list, enter the SETUP command window by using the SETUP command from the MULTI Menu. Select the second option on this menu by pressing [F2], "Maintain available applications." MULTI then displays the current list of available applications and the function keys corresponding to them.

Select the item you wish to change by pressing the corresponding function key.

The command window now shows you the three data fields corresponding to this application. You can move the cursor between fields by pressing [↑] and [↓]. To enter data in any field, type the data and press [RETURN].

The first field, "Name of Application," allows you to specify up to 24 characters of a name by which the application is to be known, such as "AlphaWRITE" or "Accounts Payable."

The second field, "Default Memory Size," allows you to specify the memory size to be allocated when the operator selects the default memory size option. Specify this field as the number of bytes to be allocated. Remember you must allocate 2048 bytes or more. You can include the multiplier of kilobytes or megabytes by typing a K or M at the end of the field (e.g., 64K or 1M).

The third field allows you to specify the command line sent to the window to start it. This command line can be up to 60 characters long and will automatically have a carriage-return appended to it when it is sent to the window. By leaving this field blank you will force the window to be started at AMOS command level.

Once you have set up all three fields as desired, press MENU to return to the available applications specification menu. If you wish to define additional applications you may do so at this time. Once all applications have been defined, press MENU again to return to the SETUP command menu. You may now save your changes (by selecting the F1 option) or can return to the MULTI Menu by pressing MENU once again.

You can change an existing application, but you cannot delete it.

Your application specifications are not saved until you use the SETUP command to save the window configuration.

Using the SETUP Command to Define System Parameters

MULTI provides several features which are optional, depending on your particular hardware configuration, or the particular way you like things to work. These options are defined within the SETUP command by selecting the "Maintain miscellaneous system parameters" option by pressing F3.

By pressing F3 you call up a menu of items which control the functioning of MULTI's optional features. As with other menus of this type, press ↑ and ↓ to move through the various fields. Change the data in these fields by typing the desired text and pressing RETURN. When you have modified the fields to your satisfaction, press MENU to return to the SETUP command menu. You may then save your changes with the Save option, or return to the MULTI Menu by pressing MENU again.

MULTI lists five different fields to change in this menu, one having to do with local printers, one having to do with phone logs, and the other three affecting the operation of the optional automatic phone dialer.

Local Printer Available?

The answer to this query should be a Y or an N depending on whether you have a printer connected to the printer port of your terminal. Typing a Y causes MULTI to include the local printer option whenever a printer selection is called for. Typing an N prevents this option from appearing.

Maintain Single Phone Log?

The answer to this query should be a Y or an N depending on whether you want a single phone log kept, or multiple phone logs kept. Typing a Y causes MULTI to maintain a single centralized log of all calls placed to numbers that have phone logging enabled in their definition entry. Typing an N causes MULTI to maintain a separate phone log for each phone list entry that has phone logging enabled.

If you have more than one phone list in a single account, phone logging may only be used with one of the phone lists in that account. If you have more than one phone list, choose which phone list you will use logging for, and make sure all items in all other phone lists have phone logging turned off. (To turn logging on or off for a particular phone list item, use the PHONE command to see the phone list, go into the entry form for the item and answer N to "Maintain phone log?".)

Auto-Dialer Connection

If you have an auto-dialer available to you, enter the one- to six-character name of the terminal port to which your dialer is connected. You can determine the port name by using the TRMDEF command at AMOS command level.

If you do not have an auto-dialer available to you, leave this field blank.

Outside Line Access Code

If you are using an auto-dialer, and your phone system requires a special access code to be dialed to reach outside phone numbers, enter that access code in this field. If you do not require an access code, leave this field blank.

If you are not using an auto-dialer, the contents of this field are ignored.

Phone Dialer Type

If you are using an auto-dialer, enter the type of dialer you are using in this field. The following dialer is currently supported:

HA Hayes AT command set modem or compatible

If you are not using an auto-dialer, the contents of this field are ignored.

Using the SETUP Command to Expand a Phone List

The SETUP command allows you to expand an existing phone list to make room for new entries. See "Using the Phone List," in Chapter 8 for details.

Using MULTI on an AM-70 Terminal

Because the AM-70 supports color in a non-standard way ("field" color instead of "mode" color), MULTI makes special allowances for supporting the AM-70. This support is based on the standard AM-70 support TCRT codes, as defined in the *AMOS Terminal Service System User's Guide*.

Some software does not conform to the TCRT standards and thus does not work with MULTI. For AM-70 support to work, you *must* use the standard color commands TCRT -1,132 through -1,147. Any other method of color selection will not work correctly with AM-70 terminals and MULTI.

If your software does not conform to the standard, you will need to either modify your software (not always an easy task) or use the TCRT translation table capability to remap the TCRT codes.

Using MULTI on Non-Alpha Micro Terminals

MULTI has been designed to use many of the advanced features found on Alpha Micro terminals, such as multiple status lines, shiftable status lines, line drawing character set, function keys, etc. However, many redundant methods are supplied for entering commands, making the use of such a terminal optional. While you receive the best performance and display quality while using an Alpha Micro terminal, this section describes how to set up MULTI for use on other terminals.

Outlined below are the areas where you can expect to see reduced functionality when MULTI is used on a terminal which does not contain certain features, and where those features are not emulated via a terminal driver.

- MULTI uses the top status line to display the current date and time, as well as the currently selected window. This information is lost if no top status line is available.

- MULTI uses the bottom status line to display the name of the currently selected window when the MULTI menu is visible. If no bottom status line is available, this information is lost.

- MULTI uses function keys as an alternate method of entering commands. If no function keys are available, all commands must be entered by commands and two-character sequences. Function keys are translated via an optional translation table, to allow for use of terminals with different grouping and coding of function keys. A listing of the two-character codes is given in Appendix B.

- MULTI makes use of the line drawing character set to draw boxes around the menu, desk accessories and help screens. Without a line drawing character set, these items are less attractive and less comprehensible.

- MULTI makes use of color on terminals which support color, but does not require it. If color is supported, MULTI assumes there are a minimum of eight colors available, that both foreground and background colors can be selected, and that changing colors does not require a screen position.

- MULTI takes advantage of some of the advanced features present in Alpha Micro terminals, such as the draw and scroll box functions, and the block fill commands. While these features are not required to use MULTI, the lack of these features causes performance degradation due to the extra time which must be spent updating the terminal screen.

Setting Up for a Non-Alpha Micro Terminal

There are five basic steps to setting up a non-Alpha Micro terminal for use with MULTI. These steps are essentially the same as for any other application which makes use of the terminal function keys. The few exceptions are expanded on below. The five basic steps are:

- Create (or acquire) a terminal driver which *fully* conforms to the terminal driver standard which was established with AMOS 1.3. Such drivers are available for purchase from a wide variety of sources.

  The driver must support not only the full complement of TCRT codes (at least those which the terminal can handle) but must also support the TRMCHR call and have the terminal capability bits set properly. Because MULTI is very terminal intensive, it is quite susceptible to errors in this area.

- Create a function key translation table, as described in detail below.

- Enable use of the function key translation tables for use within MULTI windows.

- Define a new [SHIFT]/[MENU] key and a new SWAP WINDOWS key as described in detail below.

- Make sure the software you plan to use under MULTI fully conforms to the terminal standard. Trouble areas are likely to include programs which use features without checking to see if they are available (MULTI does not support 100% of all TCRT functions), and programs which directly output escape sequences rather than going through the terminal driver (MULTI *only* works through the terminal driver). Beware of software which purports to adhere to the terminal driver standard but in fact does not. Not only will MULTI not be able to perform correctly, but other software on your computer may also fail!

Creating a Function Key Translation Table

For MULTI to recognize the function keys on your terminal keyboard, you must supply it with a function key translation table. These tables have the extension .MLX and reside in MULTI:. The name of the file is the same as the name of the terminal driver to be used with your terminal (e.g., AM62.MLX corresponds to the AM62.TDV terminal driver).

Use the FIXTRN program to create a translation table which contains not only the function keys [F1]-[F10] (as described in Appendix B) but also maps all other function and special keys to their equivalents, such as HELP mapped to ESC-?, MENU mapped to ESC-ESC, [SHIFT]/[←] mapped to ^U, etc. (Many of the control sequences for standard editing keys are listed in the AlphaVUE QRKEY help file shipped with AMOS.) For information on using FIXTRN, see the latest *AMOS System Commands Reference Manual*.

All keys, with the exception of [SHIFT]/[MENU] and the SWAP WINDOWS keys (discussed below) have equivalent one- or two-keystroke commands. If your terminal does not have enough function keys to map all of the available commands, you need to use these multiple keystroke versions.

Enable Function Key Translation in Windows

MULTI uses the WINDOW.??X files on DSK0: to translate function keys when windows are being used. Since these files are normally configured for standard Alpha Micro terminals, you have to go through an extra step to use the proper function key translation for your non-Alpha Micro terminal.

You will load into memory correct versions of the WINDOW files without changing the original files on DSK0:. (The original files must not change on the disk because people who use MULTI on other terminals will need them.)

To find out what translation files are available for your terminal, type:

```
DIR DSK0:myterm.??X[] RETURN
```

where ***myterm*** is the name of the driver used by your terminal (for example, PCCONS.??X for users of IBM PCs). Then choose which of these you need when using MULTI. For example, you might see a list of .VUX, .WRX, and .CAX files (for AlphaVUE, AlphaWRITE, and AlphaCALC, respectively). If you plan only to use AlphaWRITE, you only need to load the .WRX file.

To load a translation file into memory (for example, PCCONS.WRX), type:

```
COPY MEM:WINDOW.WRX = DSK0:PCCONS.WRX[7,12] RETURN
```

You must have each *window* do this procedure before the window calls up a package requiring function key translation. In other words, this is not done by the master job that calls up MULTI, but by each window. The easiest way to do this is to create command files that load the appropriate translation file under the name WINDOW before calling up the software package. For example, to call up AlphaCALC for a terminal using the NEWTRM.CAX translation file, create the following command file:

```
:T
COPY MEM:WINDOW.CAX = DSK0:NEWTRM.CAX[7,13]
CALC
```

Redefining the Shifted MENU Key

In addition to those items listed above, MULTI also assumes that the SHIFT/MENU used to call up the MULTI menu generates a character with the octal value 374. This corresponds to SHIFT/F13 on Alpha Micro terminals.

If you are using a terminal which does not generate this code, or does not have an equivalent key, you need to redefine the value of SHIFT/MENU. This redefinition is done by placing a MASTERMENU command within the MULTI.INI initialization file. The format of this command is:

```
MASTERMENU=Fvalue
```

Where ***value*** is an octal number specifying the character code to be interpreted as SHIFT/MENU. This character code can be found in any ASCII character conversion chart. For example, on a PC workstation using AlphaMATE, you might want to assign SHIFT/MENU to F10. After determining that F10 generates a code of 304 octal by consulting the documentation that came with the PC, you would enter the command `MASTERMENU=F304` into your MULTI.INI file.

The F in the MASTERMENU command tells MULTI that the key you have chosen is a function key, such as generated by F1 or by using the FUNCT key. If the key you wish to use as MASTERMENU is not a function key, but is a simple control character instead, do not include the F in the MASTERMENU statement.

If your computer requires that several different terminals are to be used with MULTI, you may wish to use the terminal dependent .INI file feature to ease the task of setting up the proper MASTERMENU commands. By placing terminal dependent initiali-

zation files containing the appropriate MASTERMENU and SWAPKEY commands in MULTI:, MULTI automatically uses the correct file each time MULTI is started.

Terminal dependent .INI files have the name "term.INI" where "term" is the name of the terminal driver used by the terminal invoking MULTI. MULTI searches MULTI: for one of these files regardless of whether a MULTI.INI file has been located. The files AMWS.INI and PCCONS.INI are provided on the release tape for support of AlphaMATE and the AMPC to serve as examples of using terminal dependent .INI files.

Redefining the Swap Windows Key

MULTI defaults the Swap Windows key to a value of 263 octal, corresponding to a [FUNCT]/[S] on an AM-6x terminal.

If you are using a terminal which does not generate this code, or does not have an equivalent key, you need to redefine the value of the Swap Windows key. This redefinition is done by placing a SWAPKEY command within the MULTI.INI initialization file. The format of this command is:

```
SWAPKEY=Fvalue
```

Where ***value*** is an octal number specifying the character code to be interpreted as the Swap Windows key. This character code can be found in any ASCII character conversion chart. For example, on a Workstation using AlphaMATE, you might want to assign the Swap Windows key to [SHIFT]/[F10]. After determining the shifted [F10] key generates a code of 335 octal, you would place the command `SWAPKEY=F335` within your MULTI.INI file.

The F in the SWAPKEY command tells MULTI that the key you have chosen is a function key, such as generated by [F1] or by using the [FUNCT] key. If the key you wish to use as SWAPKEY is not a function key, but is a simple control character instead, do not include the F in the SWAPKEY statement.

If your computer requires that several different terminals are to be used with MULTI, you may wish to use the terminal dependent .INI file feature to ease the task of setting up the proper SWAPKEY commands. By placing terminal dependent initialization files containing the appropriate MASTERMENU and SWAPKEY commands in MULTI:, MULTI automatically uses the correct file each time MULTI is started.

Terminal dependent .INI files have the name "term.INI" where "term" is the name of the terminal driver used by the terminal invoking MULTI. MULTI searches MULTI: for one of these files regardless of whether a MULTI.INI file has been located. The

files AMWS.INI and PCCONS.INI are provided on the release tape for support of AlphaMATE and the AMPC to serve as examples of using terminal dependent .INI files.

Redefining the Select Window Key

MULTI defaults the Select Window key to a value of 272 octal, corresponding to a [FUNCT]/[Z] on an AM-6x terminal.

If you are using a terminal which does not generate this code, or does not have an equivalent key, you need to redefine the value of the Select Window key. This redefinition is done by placing a SELECTKEY command within the MULTI.INI initialization file. The format of this command is:

```
SELECTKEY=Fvalue
```

Where ***value*** is an octal number specifying the character code to be interpreted as the Select Window key. This character code can be found in any ASCII character conversion chart. For example, on a Workstation using AlphaMATE, you might want to assign the Select Window key to [SHIFT]/[F10]. After determining that [SHIFT]/[F10] generates a code of 335 octal, you would place the command `SELECTKEY=F335` within your MULTI.INI file.

The F in the SELECTKEY command tells MULTI that the key you have chosen is a function key, such as generated by [F1] or by using the [FUNCT] key. If the key you wish to use as SELECTKEY is not a function key, but is a simple control character instead, do not include the F in the SELECTKEY statement.

If your computer requires that several different terminals are to be used with MULTI, you may wish to use the terminal dependent .INI file feature to ease the task of setting up the proper SELECTKEY commands. By placing terminal dependent initialization files containing the appropriate MASTERMENU, SWAPKEY, and SELECTKEY commands in MULTI:, MULTI automatically uses the correct file each time MULTI is started.

Terminal dependent .INI files have the name "term.INI" where "term" is the name of the terminal driver used by the terminal invoking MULTI. MULTI searches MULTI: for one of these files regardless of whether a MULTI.INI file has been located. The files AMWS.INI and PCCONS.INI are provided on the release tape for support of AlphaMATE and the AMPC to serve as examples of using terminal dependent .INI files.

Disabling Split Screen

MULTI emulates the split screen function on terminals that have "scroll box" capability. This emulation of split screen is important for support of early versions of AlphaWRITE, such as version 1.2A. Without split screen emulation, there would be no "ruler" displayed within AlphaWRITE.

However, newer versions of AlphaWRITE, such as 2.1C, are capable of performing their own split screen emulation. Because AlphaWRITE knows exactly what it needs to do, it can perform a better emulation than MULTI. While MULTI's emulation works correctly, it may be less visually attractive, particularly when used on older terminals, such as the AM-62A.

For this reason, we recommend that you disable split screen emulation in MULTI when using the combination of modern AlphaWRITE (2.1 or later) and an AM-62 or AM-62A terminal. You may do so by including a SPLITSCREEN command to the MULTI.INI initialization file. The format of this command is:

```
SPLITSCREEN=value
```

where ***value*** is either `TRUE` or `FALSE`. MULTI defaults to a value of ***TRUE***, where MULTI emulates split screen itself. Selecting a value of ***FALSE*** disables split screen emulation in MULTI.

Using AlphaOFFICE Mode

In addition to its normal appearance, MULTI can be made to emulate the user interface of the AlphaOFFICE product. To do so, add the OFFICE command to your MULTI.INI initialization file. The format of this command is:

```
OFFICE=TRUE
```

When this command is present in your MULTI.INI file, MULTI will take on the appearance of the AlphaOFFICE product.

Modifying the Directory Accessory Extension List

The DIRECTORY desk accessory displays a file description based on the extension of the file. The cross-reference from extension to description is contained in the file DSK0:EXTDEF.SYS[7,0].

To add additional extensions and descriptions to this file, or to modify existing descriptions, you can edit and reassemble the file EXTDEF.M68, which is distributed with MULTI in account [10,2]. Reassembling this file creates the file EXTDEF.SYS, which must then be copied into DSK0:[7,0].

Modifying the Message and Menu Files

MULTI stores all of its messages and menus in two external files, MLTMSG.lan and MLTMNU.lan, where "lan" is the language dependent file extension for the language selected for your job. For example, the American English language files as MLTMSG.USA and MLTMNU.USA. These files are stored in DSK0:[7,15] or MULTI:.

MULTI always looks for these files using the language dependent extension specified in the language definition table currently selected by your job.

By editing the source files MLTMSG.M68 and MLTMNF.M68, you can change the messages and menus to reflect your own language preference.

System Performance Considerations

When doing capacity planning for a computer where MULTI is to be used, it is important to realize that each window opened is effectively another "user" on the computer. Each window requires the same system resources (memory, CPU cycles, etc.) as if you added another terminal and another human being to use it.

Since MULTI allows each user to keep several different tasks running at the same time, it is important to look more at the number of *windows* in use on the computer, rather than the number of *people* using the computer, when evaluating system requirements.

MULTI makes use of many different overlays to reduce its resident memory requirements. These overlays are located in DSK0:[7,15] (or MULTI:) and all have a name of the form "MLTnnn.OVR", where "nnn" is a three digit number. However, while this does reduce memory requirements, it imposes a slight performance penalty since it must fetch these overlays off the disk when they are needed. Because all of these overlays are both re-entrant and re-usable, they can be loaded into either system or user memory to increase performance. On those computers making use of disk caching, it may be more practical to lock these overlays into the cache, rather than loading them into memory.

Restricted Programs

Because MULTI actually operates as several jobs on your computer, there are certain programs which, because they require that only one job be running on the computer, cannot be used within MULTI. Any program which performs a JLOCK call to restrict the computer to a single job, and then expects to be able to perform terminal input or output will not function under MULTI. Examples of this are the VCR backup and restore program (except when the `/TRANSFER` option has been specified), and the DSKANA program.

In addition, because MULTI allocates memory to your windows itself, programs which directly manipulate the assignment of memory on the computer (such as the MEMORY program) may not be used within MULTI.

Programs which directly access the hardware or have critical timing constraints, such as disk formatting programs, also may not function.

APPENDIX B

UNIVERSAL KEY SEQUENCES

Key Equivalents

This appendix describes the optional character codes which may be substituted for function keys during the operation of MULTI on non-Alpha Micro terminals.

Function Key Name	Universal Character Sequence
F1	Escape A
F2	Escape B
F3	Escape C
F4	Escape D
F5	Escape E
F6	Escape F
F7	Escape G
F8	Escape H
F9	Escape I
F10	Escape J
F13 (MENU)	Escape Escape
Shifted F1	Escape K
Shifted F2	Escape L
Shifted F3	Escape M
Shifted F4	Escape N
Shifted F5	Escape O
Shifted F6	Escape P
Shifted F7	Escape Q
Shifted F8	Escape R
Shifted F9	Escape S
Shifted F10	Escape T
Shifted MENU	octal 374
Swap Windows	octal 263
Select Windows	octal 272
Help	Escape ?

For character codes that can be substituted for commonly used editing keys when using MULTI's desk accessories (e.g., CTRL/T for NEXT SCREEN), see the AlphaVUE help file shipped with AMOS.

APPENDIX C

ADAPTING SOFTWARE TO THE MULTI ENVIRONMENT

Introduction

While one of the most important features of MULTI is that it does not require any changes to existing software, MULTI offers a number of features you may wish your software to take advantage of.

This appendix describes some of those features and the programming techniques required to use those features.

Determining the Controlling Job

MULTI operates by *spawning* new jobs on the system and controlling them from the job which is running the MULTI program. Each of these spawned jobs is a complete AMOS job, with all capabilities of any other job on the system. It is attached to a terminal (albeit a pseudo terminal) and can perform terminal input and output just as any other job would. Disk I/O is also unaffected by being performed by a spawned job.

Because the jobs spawned by MULTI are so much like any other job, the operation of most programs is completely transparent and there is no need for a program to be aware it is being run under the control of MULTI. There are times, however, when it is useful for a spawned job to be able to determine which job is controlling it. For example, an application which generates filenames based on the job running the application might want to use the name of the controlling job, rather than the one assigned by MULTI when the job is spawned.

To allow a program to determine the controlling job, the JOBATT field within the spawned job's Job Control Block (JCB) is set to contain the address of the controlling job's JCB. By using this pointer, a program can have full access to information regarding the controlling job.

Remember that a spawned job can in turn spawn other jobs. Thus it may be necessary to follow the JOBATT links through multiple jobs to determine the ultimate parent job.

Preventing Context Switching Under MULTI

Some special operations require that a program prevent context switching during critical time periods. This is often accomplished by the use of the JLOCK monitor call. When used for short durations, with no terminal I/O being done during the locked time, this is compatible with MULTI.

However, in circumstances where the duration of the lock is lengthy (such as during a backup operation) or where terminal I/O must be performed, use of JLOCK under MULTI will cause a system hangup. This is because in locking out all other jobs, MULTI itself is locked out, and can no longer service its spawned jobs.

Under AMOS 2.0 and later a new monitor call, PLOCK, has been provided to allow locking under MULTI and other programs which spawn jobs. This call, instead of locking out all other jobs, locks out only those not "related" to the job issuing the PLOCK. This allows MULTI to continue to run and service its spawned jobs without interference. For this reason, the use of PLOCK is generally preferable to JLOCK.

For More Information

Additional information on the Job Control Block, JOBATT, the TRMCHR call, and the PLOCK call, can be found in the *AMOS Monitor Calls Manual*.

APPENDIX D

TERMINAL EMULATION DESCRIPTION

Introduction

MULTI performs most of its functions through its ability to emulate a terminal screen, or more importantly, several terminal screens at a time. While a great deal of effort has been expended to fully emulate all required TCRT (terminal command) codes, there are some areas which may cause some difficulties with certain software packages.

MULTI was written to adhere to the terminal usage standards which were worked out by Alpha Micro, in conjunction with its dealers and users. All software which likewise adheres to these standards will function properly under MULTI. However, there does exist software which does not adhere to these standards. While MULTI as much as possible has been designed to accept this software, there exist some possible areas of incompatibility with this non-standard software.

This appendix describes these areas and possible difficulties you might have.

Attributes

MULTI supports either mode or field screen attributes, depending on the type of attributes supported by your physical terminal.

Save Screen Area Commands

MULTI supports the save and restore screen area commands (TCRT -1,148 and -1,149), regardless of whether your physical terminal has support for these commands. This means that programs running under MULTI have access to these commands on all terminals, rather than just on the newest terminals.

Split Screen Support

MULTI 2.1 does not support or emulate split screen functions. Programs, such as AlphaWRITE, which make use of split screen must perform appropriate emulation themselves.

Local Printer and Transparent Print Support

The transparent print and print screen functions (TCRT -1,79; -1,82; and -1,83) require special support within MULTI. When these print functions are actually in progress, MULTI does not recognize the shifted [MENU], SELECT WINDOW or SWAP WINDOWS keys, thereby avoiding corrupting the printer output with MULTI's own output.

If a job requests transparent print or print screen when another window is selected, the requesting job is put into a wait state until it is selected; no printing occurs until it is selected.

You are informed that a window is waiting to use the local printer any time you call up the MULTI Menu: a short message displays on the bottom status line. The SELECT menu also displays a blinking P next to the name of any window which is waiting for the printer.

By implementing this system of waiting for the availability of the local printer, MULTI allows completely transparent use of the local printer while still providing its many benefits.

Color Terminals

MULTI fully supports the use of color terminals which adhere to the TCRT -2 and -3 color specification standard. Such terminals include the PC workstations used with AM-PC, AlphaLAN, inFront, and STEP, as well as terminals like the AM-72 and AM-75.

When used with a color terminal, MULTI not only supports applications which use color, but also uses color itself: using different colored backgrounds to help distinguish one menu from another.

132 Column Mode

Some terminals, such as the AM-65, support the use of an optional 132-column mode, allowing greater amounts of data to display on the terminal. MULTI supports this feature and provides full access to selectable 80 or 132 column modes.

APPENDIX E

JOB SCHEDULING UNDER MULTI

Introduction

MULTI makes use of *spawned* jobs to allow it to perform its functions. Because these spawned jobs are no different from any other AMOS job, they behave much the same as any other job. Because they operate under control of another job (MULTI), however, they behave a bit differently. This appendix explains those differences.

Job Scheduling

One of the advantages of MULTI is its ability to support true *multi-tasking*: the ability to have several tasks active at once, all performing work. By spawning multiple jobs, MULTI assigns one job per task, using AMOS's timesharing capabilities to make sure each gets its fair share of time by controlling when and for how long each job runs.

To maximize the benefit of this multi-tasking, MULTI never suspends a job. If the job is CPU bound it continues to execute regardless of what you are doing in MULTI (desk accessories, menu, alarm clock, etc.). If the job is doing terminal output, then it suspends itself when its output buffer fills up. It is one of MULTI's duties to empty the output buffer and update the screen. Normally MULTI is doing this automatically in the background while performing its other tasks. The only time MULTI stops doing this, and therefore allows its other jobs to stay in terminal output wait state, is when you are using the MULTI Menu or one of the desk accessories. If you are directly using a window, all jobs are being serviced for terminal output.

The reason jobs are allowed to stay in terminal output state when a desk accessory or the MULTI Menu is being used is to allow the cursor to stay located in the desk accessory. Early versions of MULTI which continued to update the screen when a desk accessory was being used were deemed "too distracting" by the people using them at the time. Thus the problem is more one of aesthetics than a technical one.

Job Characteristics

Each job on the system has certain characteristics and settings associated with it. These characteristics control such things as job priority, output radix, whether device error messages are displayed, etc. When MULTI spawns a new job to assign to one of your windows, it gives that new job the same characteristics and settings as your original job (the one that called up MULTI), giving the window the same properties as your original job.

If you want to change these settings, the SET program functions normally within each window to allow you to specify individual characteristics for each window.

Job Names

When MULTI creates a task to run in a window it must assign it a name. AMOS requires that all jobs on the system have a unique name. To ensure it is a unique name, and to identify it as a spawned job, MULTI always gives its jobs a name of the form "TSKxxx" where "xxx" is three alphabetic characters.

Because MULTI is controlling these jobs it is important they not be interfered with by other jobs or programs on the system. For this reason, most utility programs on the system do not allow themselves to be used with spawned jobs. For example, the FORCE and KILL programs may not be used on spawned jobs.

Resetting Jobs

The use of programs which "reset" jobs to AMOS command level, while never recommended, are particularly dangerous to use on a job running MULTI. Use of such a program almost invariably causes system failure, including crashing and the corruption of disk files.

APPENDIX F

USING SCRIPTS UNDER MULTI

Introduction

When demonstrating MULTI, and applications that run under MULTI, it can be useful to set up a script file that allows unattended continuous operation of MULTI; MULTI's script file capability allows you to do this.

Defining and Using Scripts

Script files are ASCII text files (created with AlphaVUE) with the extension .WSF. These files are very similar in structure to task manager control files. After creating a script file, you invoke MULTI with the command:

`MULTI/AUTO:scriptfile` RETURN

where ***scriptfile*** is the name of the script file you want to use. Script files are simply text files that contain the list of commands to be sent to MULTI. Any text included in the script file is sent to MULTI, just as if you had entered it on the keyboard.

When MULTI reaches the end of a script file, it automatically starts over again at the beginning, allowing for continuous demonstrations.

All MULTI script files must begin with a blank line or the first command in the script file will be ignored.

Embedded Script File Commands

In addition to the standard text, MULTI also supports special commands within script files. All of these commands must be placed on their own separate line, and begin with a dollar sign.

The `$MENU` command is the equivalent of pressing the shifted MENU key, regardless of what character the MENU key is defined as.

The `$SLEEP` command allows you to specify that MULTI should pause before continuing. This command takes a single integer argument which is the number of seconds to pause.

The $^ command allows you to specify control characters to be sent to MULTI. Any characters following the caret are converted to their control equivalent before being used by MULTI.

The $+ command allows you to specify characters to be sent to MULTI without a terminating carriage return.

Sample Script File

The sample script file shown below performs a simple demonstration of MULTI's desk accessories. The following sample script file is shipped with MULTI as DSKACC.WSF, and is called up by typing:

MULTI/AUTO:DSKACC/NOSTART RETURN

This is what the sample script file looks like:

```
$SLEEP 5
CALCULATOR
$SLEEP 1
1633.12
$SLEEP 1
6153.65
$SLEEP 1
32.09
$SLEEP 1
$MENU
CALENDAR
$^^
$SLEEP 1
$^J
$SLEEP 1
$^J
$^J
$SLEEP 1
$^H
$SLEEP 1
$^H
$SLEEP 1
$^T
$^T
$SLEEP 1
$^T
$^T
$SLEEP 1
$MENU
$SLEEP 1
NOTE
$SLEEP 3
$MENU
ALARM
```

```
$SLEEP 2
$MENU
PRINT
$SLEEP 3
$MENU
DIR
$SLEEP 1
$^T
$SLEEP 1
$^R
$SLEEP 1
$MENU
```

Hints for Using Scripts

- The ^[characters represent an ESCAPE, so to include an ESCAPE A (for example) in your script file, use the lines:

  ```
  $^[
  $+A
  ```

- When demonstrating applications, be sure to include adequate `$SLEEP`commands to give the application time to work; otherwise, some script commands may be lost.

- When looping scripts, be sure to include a `$SLEEP` command at a point where the main menu is visible and it is possible to quit MULTI so the user can exit.

MULTI User's Manual
Document History

REVISION	DATE	CHANGE
A00/A01	March 1986	First release. Documents MULTI Version 1.0. Issued under part number DSS-10173-00.
00	August 1989	Issued under new part number, DSO-00083-00. Rewritten to document MULTI version 2.0.
01	January 1994	Updated to reflect MULTI version 2.1. °

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